

Data Governance & Operating Model Roadmap

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Executive Summary



Context & Drivers for Change

In response to increasing data governance expectations, recent data audit findings, growing demand for analytics, and the University's digital and AI ambitions, Sia Partners was engaged to support the uplift of the University's data operating model and governance.

Context

UWA's increasing reliance on data to support student outcomes, regulatory obligations, and strategic decision-making has heightened expectations around data quality, consistency, and trust. These expectations, however, are not being consistently met resulting in growing frustration across the business and reduced confidence in data-driven insights.

Recent data governance audit findings confirmed these challenges and recommended the introduction of a formal, enterprise-wide data operating model to address some of the systemic issues the university faces.

In response, UWA's IT division commissioned this project to assess its current data operating model and governance, and to define a target state data operating model at a high level. This work used a single use case (student onboarding & enrolment) to anchor findings and recommendations but was designed so the foundations are established to help scale the model to meet the diverse needs across the university.

The outcome will provide a practical foundation to support future technology strategy, inform investment decisions, and enable sustainable improvement in the University's data capability.

External drivers



Regulatory requirements

Increasing government, industry and community expectations that require clearer data ownership, traceability, and governance controls.



Competitive pressure

Strengthening enterprise data foundations is a critical enabler of advanced analytics, AI, and automation, creating a competitive imperative to modernise data capability.

Internal drivers



Loss of data trust and quality

Persistent gaps in central data services have driven the development of local, ungoverned 'shadow' data solutions across the university, contributing to a cycle of poor quality data and confusion.



AI strategy

UWA is currently developing its AI strategy, which will require significantly stronger data quality, standards, and accountability to be successful.



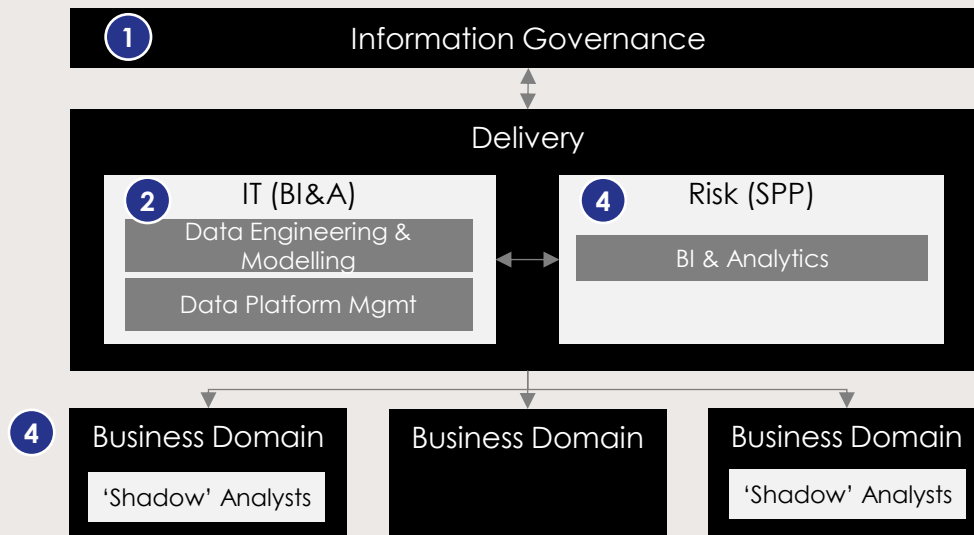
Existing momentum

UWA's central data teams are working to address the businesses' concerns through a range of uplift initiatives (e.g. EDW migration), but their effort may stall without data governance foundations in place – clearer roles, stronger capability and enterprise-wide alignment.

Current State Findings Summary

UWA's currently has an informal, semi-centralised data operating model that lacks the data governance frameworks, capability and clarity of responsibilities required for the university to have consistent access to quality and valuable data and insights.

3 Current State Operating Model



UWA currently operates a **semi-centralised data operating model**. IT (BI&A) builds and maintains verified enterprise data models, a central reporting team (SPP) produces most institutional reporting, and business domains primarily consume outputs with limited (formal) ability or accountability to manage their own data products. This set up is compliance focused, without grounding in data governance frameworks.

Key current state issues / gaps:

1 Data Governance vs. Information Governance

UWA has established Information Governance practices that manage privacy, risk, and compliance obligations; however, complementary Data Governance frameworks / capability are required to define the data ownership, quality expectations, and controls needed for consistent enterprise reporting and analytics. For data governance tools that do exist (e.g. Purview), there is limited awareness of their existence/value.

2 Central Data Team Overloaded

The central IT data team (Business Intelligence & Analytics) has been the default team expected to fill the university's data governance gaps, on top of their BAU work. This important function is now stretched them across too many responsibilities, from technical delivery to informal governance decisions and broader business education, creating a bottleneck felt across the university.

3 Unclear Operating Model

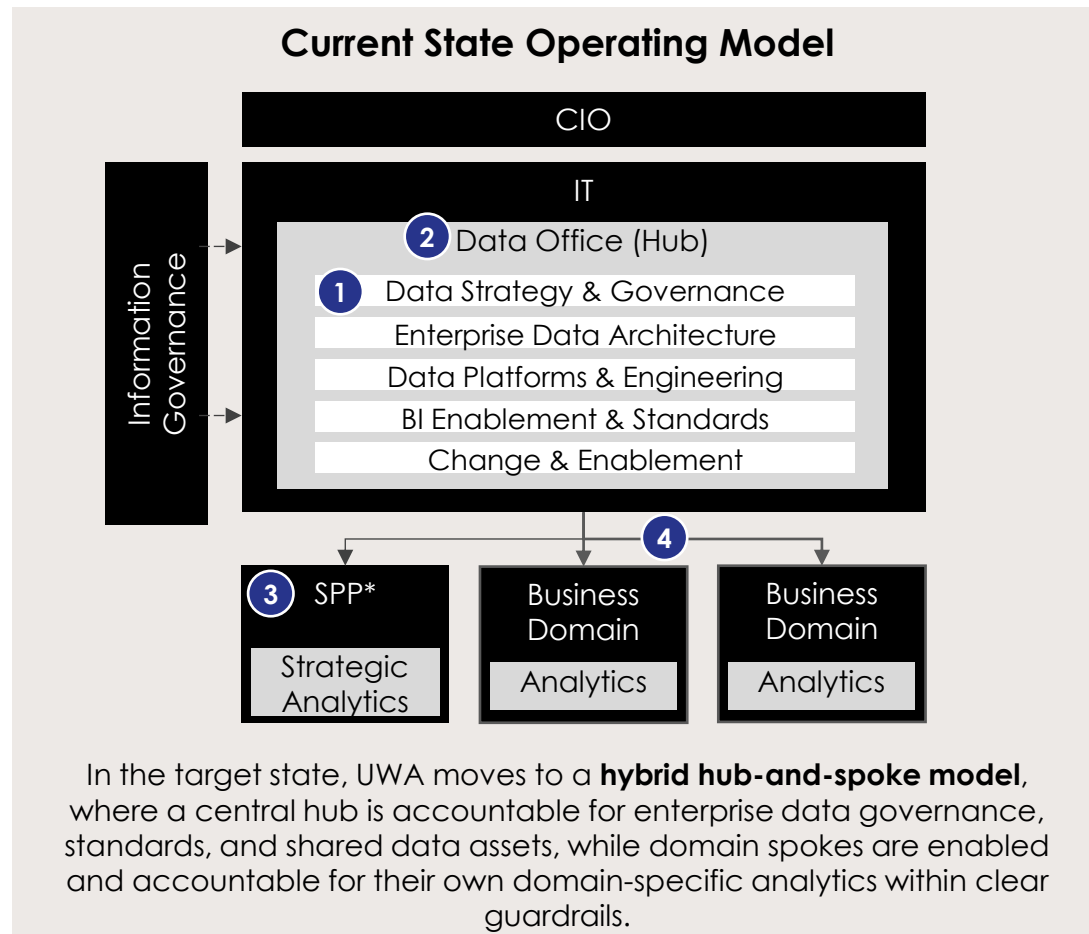
The absence of a clearly defined data operating model outlining accountabilities and interactions between the central IT data team (BI&A), the central reporting function (SPP), and business domains has driven confusion on 'who does what' and is a key contributor to poor data outcomes. This fragmentation spans the full data lifecycle, from how demand is captured and delivered, to how data issues and feedback are escalated.

4 Shadow Analytics Without Guardrails

There is no clear owner for delivering domain-specific insights at UWA, and no agreed analytics guardrails. As a result, analytics work has drifted to wherever capability exist, often pulling the central strategic analytics team (SPP) into a routine dashboarding service provider, while 'shadow' analytics teams also emerge in high-needs business areas. Without clear guardrails and (often) the appropriate capability, analytics produced by these teams are often conflicting and damaging to the efforts of the central data team to develop certified datasets.

Target State Summary

In the proposed target data operating model, UWA will have a Data Office ("Hub") that provides centralized governance and platforms, while business domains ("Spokes") build their own analytics using these shared foundations, each with the level of support and oversight adjusted based on each domain's readiness and capability.



Key characteristics

1 Formal Data Governance

A new model introduces a formal data governance framework to define ownership, quality expectations, and controls across enterprise and domain data. Dedicated data governance roles will be introduced to design, promote and manage this framework.

2 Data Office (IT) Rebranded

IT's central data team is rebranded as the Data Office (i.e. Hub) - a central data asset provider (i.e. platforms and enterprise datasets) and standards keeper that enables each 'spoke' to deliver their own domain-specific insights in a controlled, streamlined setting. This rebrand requires new roles and clearer responsibility boundaries within the team to eliminate confusion and reduce demand bottlenecks.

3 Analytics Capability Distributed to Domains

Domains operate as "spokes" with embedded analytics capability that is supported by the Data Office. This allows teams to build and own their own domain-specific insights using governed, certified data assets from the Data Office, reducing wait times and enabling faster decision-making. SPP will focus on strategic insights, rather than being a central analytics function for all Domains.

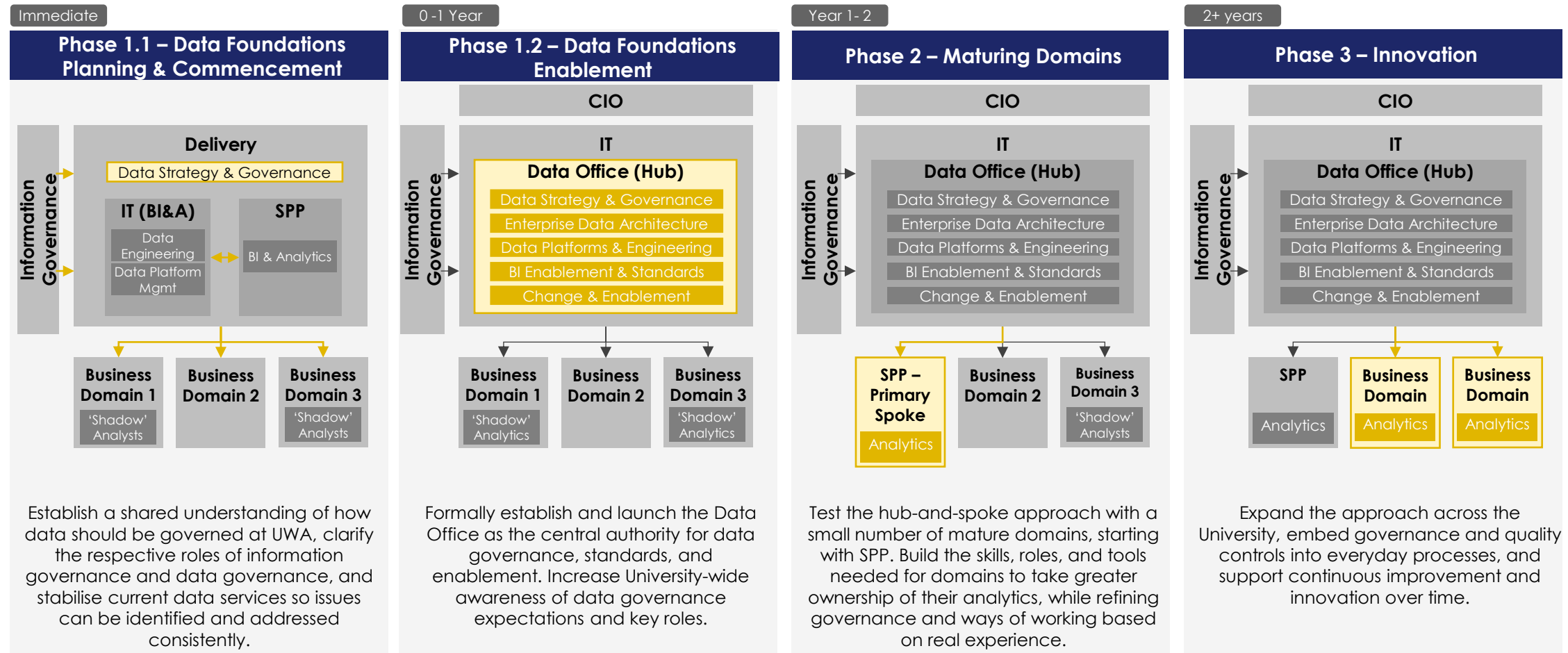
4 Clearer Responsibilities & Collaboration

The introduction of clearer ownership, defined escalation paths, and transparent governance forums will ensure data quality issues are surfaced and resolved rather than worked around. Users have a clear route to raise concerns and get support, eliminating the need for workarounds and shadow solutions that undermine consistency and trust.

*SPP is the proposed first spoke to implement because it already works relatively closely with IT, has the maturity to adopt new standards, and can demonstrate the value of the model while delivering immediate strategic impact.

Proposed Transitionary phases

The proposed roadmap reflects a staged path – initially focused building the ‘hub’ with the appropriate standards, processes, roles and skills. From there, each spoke will be introduced in a staged manner to minimize change impacts and ensure the operating model supports each domain’s unique needs.



1. Assess Phase

Objective

The assess phase aimed to identify the current state challenges & gaps that need to be addressed to improve the university's data governance and operating model.

Activities

Stakeholder Engagement

We have engaged ~30 key stakeholders across:

- IT Architecture, Data and Delivery teams
- Student use case representatives (i.e. Student Administration, etc)
- Information Governance
- Load Planning & Management

Document Review

We have reviewed ~20 key documents including:

- IT strategy & architecture
- Key process flows & roles & responsibilities
- Work from recent internal projects

Deliverables

- ✓ **Data Governance & Operating Model Maturity Assessment**
- ✓ **Documentation of Current State and Associated Pain Points (i.e. responsibilities, roles, structure, processes)**



Current State Assessment Summary

The Assess phase was used to understand how UWA currently manages, governs and delivers data / insights. This includes an assessment of structure, roles, processes, culture and systems. The Assess Phase findings inform the target state operating model transition.

Symptoms



Inconsistent, unreliable information

Different teams produce different numbers for the same metric, and data issues appear late, creating uncertainty and slowing decision-making.



Low trust leading to fragmented reporting

Users bypass central datasets, rely on local extracts or spreadsheets, and create parallel reporting streams when central logic doesn't meet their needs.



High effort to produce and maintain insights

Frequent breakages, unclear ownership, and duplicated work result in significant manual validation, rework, and support effort for enterprise data teams.

Interview

Analyse

Validate

Themes & Key Findings

1. Governance

Data Governance is conducted in an ad-hoc fashion, with the university oriented toward privacy and compliance only rather than data quality and value. There are no roles and limited rules and tools dedicated to data governance practices.

2. Data Lifecycle

Enterprise-wide standards for data capture, validation, business rules, quality management, and definition governance are non-existent, underdeveloped or inconsistently applied.

3. Capacity & Capability

Key data governance roles and data delivery capability is missing or thinly spread. Poor data literacy within the business leads to inconsistent modelling and duplicated logic.

4. Demand Management

Intake and prioritisation of data support requests aren't well-defined and/or communicated and the central data team's role isn't well defined causing confusion and backlog across the business.

5. Tools & Platforms

Core platform capabilities (lineage, cataloguing, automation) and shared datasets aren't fully operationalised, limiting reuse, consistency, and monitoring.

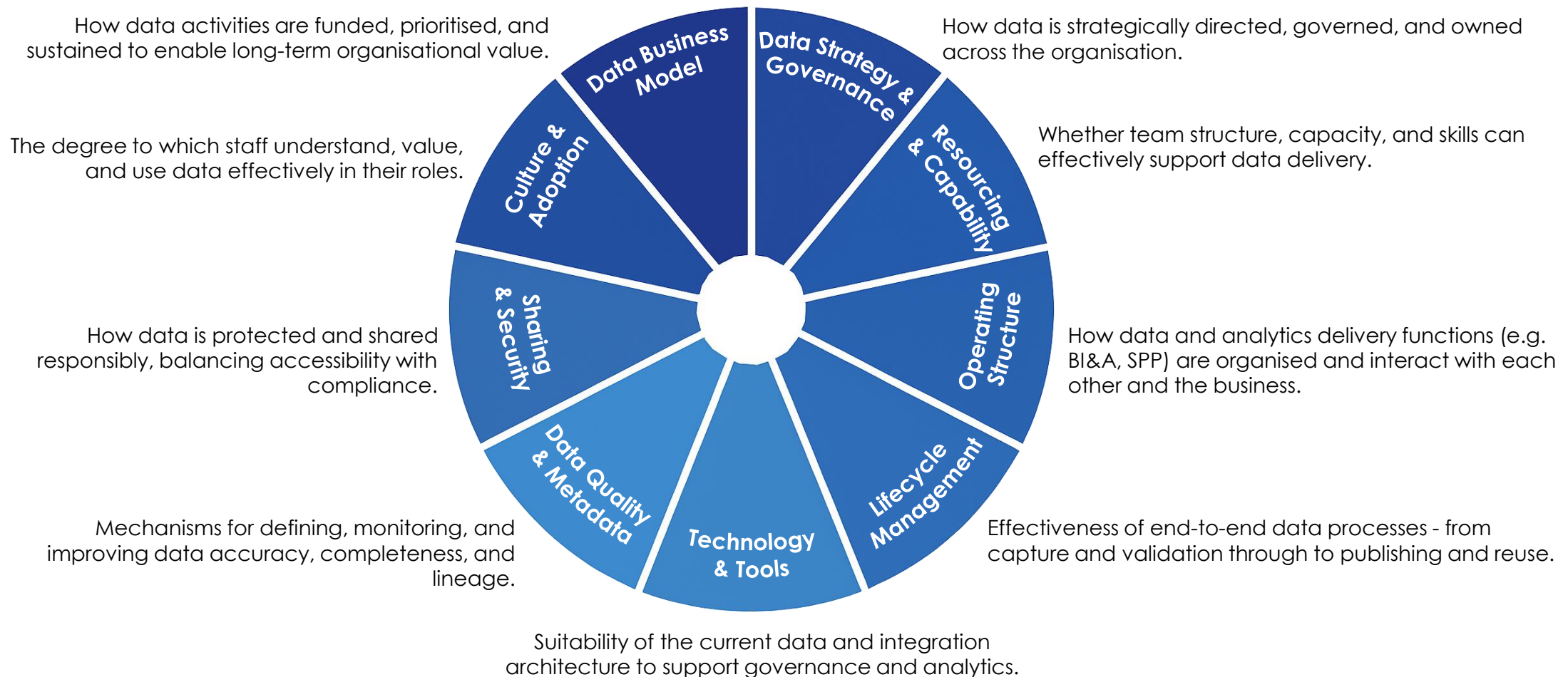
6. Funding & Value

Funding is split across functions without a coordinated investment plan, and success measures focus on technical activity rather than business value or maturity.

Improvement Initiatives

Data Governance & Operating Framework Assessment

To diagnose the causes of UWA's data challenges, we assessed across nine interdependent dimensions spanning Data Governance and the broader data operating model. These pillars summarise the assessment areas that will inform the Data Governance and Operating Model Roadmap.



Assessment Results

Assessment Criteria

	Level 1 - Exploratory	Level 2 - Experimental	Level 3 - Operational	Level 4 - Optimized	Level 5 - Transformed	Assessment Comments
Data Strategy & Governance How data is strategically directed, governed, and owned across the organisation.		◆				- No endorsed enterprise Data Strategy aligned to UWA 2030. - Some governance roles exist but limited actionability, awareness and not aligned to data. - Decision rights unclear; accountability falls disproportionately on IM/IT. - Data governance not yet embedded in BAU or organisational leadership.
Data Business Model How data activities are funded, prioritised, and sustained to enable long-term organisational value.		◆				- No data funding model or investment plan currently (Roadmap to support). - Data initiatives are supported through BAU resource allocations or ad hoc project funding from IT or individual faculties. Some cases using external resourcing. - No value measurement framework and a limited central prioritisation process.
Resourcing & Capability Whether team structure, capacity, and skills can effectively support data delivery.			◆			- BI/IT have core skills; domains have minimal analytics or data stewardship capacity. - Data roles documented but have limited embedment into role definitions. - No enterprise-wide literacy or skills uplift program. - Capability is uneven and centrally concentrated within IM/IT teams.
Operating Structure How data and analytics delivery functions (e.g. BI&A, SPP) are organised and interact with each other and the business.			◆			- Data Operating model is largely centralized, with decentralized 'shadow' analytics capability across the org. - Use case intake and prioritisation inconsistent across domains. - BI supports demand rather than enabling domain ownership – reactive, demand driven. - Centralised - No data producer/consumer operating model – no domain ownership.
Lifecycle Management Effectiveness of end-to-end data processes - from capture and validation through to publishing and reuse.			◆			- Data capture processes vary by system (Callista, CRM, LMS) with limited alignment or any governance considerations. - Change control, validation, and limited reconciliations inconsistent and often manual. - No cross-system lifecycle view or standardised processes for reuse. - Data lineage tracking is limited.
Technology & Tools Suitability of the current data and integration architecture to support governance and analytics.				◆		- Azure-based platform functional but not business-led. - DW covers core domains but with inconsistent business usage. - No metadata catalogue, lineage tooling, or API governance. - Limited architectural standardisation across domains.
Data Quality & Metadata Mechanisms for defining, monitoring, and improving data accuracy, completeness, and lineage.		◆				- No enterprise DQ framework, tooling, or accountability. - Metadata not captured; definitions inconsistent across systems. - No catalogue; lineage limited. - Reactive issue management led by BI/IT only – some reporting on data quality.
Sharing & Security How data is protected and shared responsibly, balancing accessibility with compliance.				◆		- PII processes, consent, archiving and security controls established, however information management led. - Compliance frameworks in place via IG and Security. - Cross-system sharing fragmented; access models vary by platform.
Culture & Adoption The degree to which staff understand, value, and use data effectively in their roles.			◆			- Data culture exists mainly in technical teams. - No organisation-wide literacy or data enablement program. - Information Stewardship culture available but not active; low awareness of data value.

1. Data Governance - Findings

While UWA has an Information Governance Framework, an active IG team, the IS&GSC committee, and named Information Custodians and Stewards, governance remains compliance-focused and is not operationalised for data governance. UWA lacks dedicated data governance policies, standards, procedures, and an operating model to support data quality, ownership, business rules, lifecycle management, and value creation.

ID	Finding Title	Detail	Roadmap Considerations
DGF1	Data governance – op model not defined or embedded	- No clearly established data governance operating model to support data management items such as data quality, data definitions, lineage, or rule ownership. - IG function focuses on compliance (FOI, privacy, records management) rather than data management, quality and value. - No clear definition and role of Information Governance vs Data Governance. - Information governance roles exist which do contain reference to data quality management, but operationalization of such roles have been ad hoc and inconsistent.	- Development of Data Governance Framework – interoperability of IG function & data governance op model and guidelines. - Development of Data Governance Op Model – aligned with archetypes and staged models. - Data Product Operating Model – aligned with Op model. - Training & Enablement - Alignment on Policies, Standards & Guidelines
DGF2	Data governance - policies & standards non-existent	- No specific data governance standards, policies or guidelines. - Limited data quality rules, expectations or enablement and ongoing data quality resolution. - Limited and solid Business Glossary exists only narratively - not mapped to model fields or CDS logic. - Limited standards for metadata/lineage, defect lifecycle, dataset retention, or reference/master data management.	- Development of Data Guidelines, Standards & Policies across – DQ, MDM, Meta-Data and more. - Development of foundational data governance & management items – CDE's, business glossary, data dictionary, lineage, DQ framework, MDM & Reference data registers, Meta-data, RACI
DGF3	Data Governance – Roles, responsibilities, governance forums, and the stewardship network are either inactive, ineffective, or not yet established.	- Data governance roles, including stewardship and ownership, are not formally defined or assigned. Responsibilities default to IG compliance functions, and expectations for data management are unclear due to missing standards, supporting documents, and governance tools. - Information Custodians/Stewards receive no / limited training and are largely unaware of their role; responsibilities not embedded in PDs/KPIs. - No focused data governance forums for specific data management activities such as data rule changes, data quality issues, definition disputes, or lifecycle decisions.	- Define data governance roles aligned to the framework and integrated with Information Governance compliance. - Refresh Information Governance roles to explicitly include data governance responsibilities and introduce additional roles where needed to support data management. - Define training requirements on updated roles and responsibilities. - Define or establish archetype relevant data governance forums.

2. Data Lifecycle Management - Findings

While BI&A publishes certified datasets using medallion architecture and version control, enterprise lifecycle standards for capture, validation, change management, rule governance, and quality are missing or inconsistent.

ID	Finding Title	Detail	Roadmap Considerations
DLM1	Data Lifecycle Management – No supporting frameworks within data lifecycle development, productionisation & continuous improvement	- While UWA is modernising its data platform (i.e. introducing a medallion architecture), the governance needed to control the data product lifecycle (e.g. definitions, business rules, certification, versioning and change management) remains under-developed. - As a result, “Gold/Certified” datasets are not anchored in enterprise-wide standards and are understood only within BI&A. Without shared rules or visibility, domains often misinterpret or misuse these datasets, reducing their reliability and value.	Design and implement a governed data product lifecycle framework, including certification, versioning, change management, communication standards, and deprecation rules.
DLM2	Data Lifecycle Management – No enterprise data capture, storage, quality validation standards	- There are no clear shared standards for data quality across systems or stages of the data lifecycle. - Different systems apply different freshness and accuracy expectations (e.g., some update daily, others weekly), and many checks are manual rather than automated. - There are also no agreed thresholds for what “good data” means (e.g., % completeness required, acceptable error rates). - As a result, issues such as stale data, missing fields, or broken pipelines are often found late, meaning UWA staff need to spend time fixing avoidable defects instead of preventing them.	- Set enterprise DQ standards for completeness, accuracy and data freshness across key domains. - Create shared visibility of DQ issues (e.g. through domain scorecards) with clear ownership. - Embed a regular DQ review rhythm to track performance and prioritise fixes across domains.
DLM4	Data Lifecycle Management – Business glossary, data dictionary & rule governance absent	- There is no enterprise approach for defining, approving and maintaining core business definitions, data rules or metadata. As a result, terms, measures and rules differ across BI&A, SPP, IG and domains. - The Business Glossary is high-level and not linked to actual fields, datasets or rules, meaning staff do not know how definitions translate into data they use.	- Develop a central business glossary and data dictionary, linked to fields, measures, and domain rules. - Establish a stewardship forum to govern definitions and rule changes across BI&A, SPP, and IG.
DLM3	Data Lifecycle Management – Weak change management for data movement	- There is no enterprise change management framework that governs how system changes (e.g. new fields), business rule updates, and BI&A dataset changes (e.g. data lake migrations) are assessed, approved, communicated, and monitored across the Hub and domains. - For example, the transition to the medallion architecture has not been consistently communicated or coordinated, resulting in many legacy dashboards across domains that remain misaligned with the new model.	Implement an enterprise data change management framework (covering system, rule, and BI&A dataset/model changes).

3. Resourcing & Capability - Findings

While UWA has strong analytical capability within central BI&A and SPP teams, significant gaps and bottlenecks remain in data engineering capacity, operational data governance, modelling, metadata management, and stewardship activation.

ID	Finding Title	Detail	Roadmap Considerations
DRC1	Resourcing & Capability - BI&A Is Over-Extended Due to Unclear Boundaries and Missing Functions	- The BI&A team currently manages ingestion, modelling, Lakehouse migrations, pipeline fixes, documentation, platform administration, reporting model development, and operational service requests. - Because key functions such as data governance, analytics enablement, and semantic modelling standards are not formally owned elsewhere, BI&A becomes the default resolver of business logic questions, definition disputes, metadata issues, and modelling fixes. - Frequent BAU interruptions and service tickets create constant context-switching, limiting the team's ability to progress strategic initiatives (e.g., Lakehouse uplift, new domain models, TCSI ingestion, SNOW/Graph migrations). - As a result, BI&A acts as both a technical delivery team and an informal governance and modelling authority, creating dependency and reinforcing bottlenecks.	- Redistribute work by clearly defining responsibilities across Engineering, Analytics, and Governance functions. - Introduce structured intake & triage to protect engineering and modelling time. - Establish roles (or uplift existing staff) to absorb governance and modelling responsibilities currently falling to BI&A.
DRC2	Resourcing & Capability - Data governance roles and responsibilities are unclear or missing	- UWA lacks clearly defined and consistently practiced data governance roles (definitions, business rules, metadata, data quality, lineage, and lifecycle management). While Information Owners and Stewards exist on paper, many are not activated or aware of their day-to-day responsibilities. - Business teams have limited familiarity with governance practices, resulting in inconsistent definitions, frequent escalations to BI&A for basic governance questions, and avoidable data quality issues. - These gaps drive reliance on central teams for interpretation and decision-making, slowing delivery and reducing confidence in university-wide reporting.	- Establish a clear, university-wide governance framework with formal responsibilities across Owners, Stewards, BI&A, and IT. - Activate governance roles through training, playbooks, and structured forums. - Improve business literacy so teams confidently manage definitions, rules, and quality within their domains.
DRC3	Resourcing & Capability - Analytics skill gaps in modelling & semantic design	- Many analysts and business teams have limited exposure to core modelling concepts (e.g., star-schema principles, centralised logic, KPI alignment). - This leads to report-specific logic being created outside BI&A, resulting in conflicting numbers, inconsistent interpretation of metrics, and duplicated effort. - The lack of modelling standards across UWA also increases reliance on BI&A to fix or reinterpret models - further contributing to their workload.	Strengthen modelling standards, provide clear guidance and templates, and build capability across business teams to reduce reliance on BI&A for corrections and ensure consistent reporting.

4. Demand Management - Findings

While BI&A and SPP have defined delivery processes, a CDS Issue Log, a BI Business Requirements guideline, and a ServiceNow intake channel, demand management remains fragmented across multiple informal pathways, BAU capacity is routinely redirected to projects, and no joint prioritisation or structured feedback loop exists for critical data issues.

ID	Finding	Detail	Roadmap Considerations
DDM1	Demand Management - BI&A role and service offering is unclear	- There is no agreed service catalogue or formal request process for data related services. BI&A currently receive queries and requests through ServiceNow, MS Teams and email. - This reduces visibility of workload and causes confusion across the business on 'who to go to' for data related issues.	- Develop central service catalogue for BI&A and SPP. - Clarify roles and boundaries collaboratively, helping avoid misalignment or unplanned central workloads. - Validate demand capacity requirements across known channels and shadow analytics requirements. - Define single governed intake channel & unified prioritization framework – embed within existing and enforce.
DDM2	Demand Management - Unclear request triage and service expectations	- BI&A currently manage an Issue Log for certified datasets, but stakeholders lack visibility into triage logic, ranking, ETAs, and prioritisation rationale. - No SLAs have yet been defined for services/enhancements.	- Review triage and prioritisation process. - Develop SLAs associated with data services.
DDM3	Demand Management - No integrated BI&A/SPP operating rhythm	- While both teams offer enterprise data services that are strongly dependent on each other, BI&A and SPP don't have a joint roadmap of upcoming demand / activities. - Cadence between both teams is ad hoc in nature – for discussing immediate issues rather than forecast work.	- Establish a more aligned operating rhythm, including coordinated sprint or planning cycles. - Support closer collaboration on shared datasets, with BI&A guiding certified layers and SPP contributing business logic.
DDM4	Demand Management - IT project intake lacks data impact assessment	- The established IT project assessment process doesn't currently require a data impact assessment. - Migration, integration, metadata, lineage, and DQ work are therefore not explicitly scoped, risking unmanaged downstream impact.	- Integrate data governance sign-off steps into project initiation. - Introduce a simple data impact lens for migrations, integrations, and rule changes.

5. Tools & Data Assets - Findings

UWA is investing in modern data platforms (i.e. Databricks, Purview, PowerBI), but the governance and operational capabilities within these tools are not yet fully activated. A structured assessment and targeted uplift will be needed to support reusable, trusted, cross-functional data assets across the university.

ID	Finding	Detail	Roadmap Considerations
DTA1	Tools & Data Assets - Governance capabilities in Purview and Databricks are not operationalised	- Metadata, lineage, and classification features are available in Purview, but they are not consistently populated or used as part of day-to-day data processes. - The EDW Data Dictionary exists but is underused and not linked to definition or rule governance.	- Assess Purview for data governance- including stewardship, metadata maintenance, and lineage capture, seek alternate solution if required. - Align Purview (or other solution) with foundational data governance practices (definitions, DQ rules, ownership) to support target state.
DTT2	Tools & Data Assets - No enterprise MDM or conformed dimensions	- UWA has not introduced core enablers such as enterprise mastering (e.g., Person, Course, Term), reference data ownership and versioning, or basic platform observability (e.g., freshness checks, schema-change alerts). - These capabilities are essential for making data governance expectations workable in practice.	Establish foundational operational capabilities (mastering, reference data governance, survivorship rules, observability) and assign clear ownership to support day-to-day data governance.
DTT3	Tools & Data Assets - Inconsistent workspace, access, and semantic governance	- Power BI workspace governance and RLS patterns exist but are not systemically enforced via platform templates, standards, or automated controls. - Local semantic models frequently recreate central logic due to limited reusable enterprise assets.	Improve reusability and visibility of enterprise logic, helping Spokes leverage certified datasets rather than rebuilding rules locally.
DTT4	Tools & Data Assets - Existing Power BI datasets cannot be easily transitioned into a certified, reusable model	- Microsoft's platform makes it difficult to retrofit many existing Power BI datasets into a Certified Dataset (CDS) structure. - Larger datasets, heavy transformations, nested logic, and calculated columns that depend on other tables often cannot be certified without significant redesign. - As a result, it is challenging to move from locally-built reports to a shared, reusable, enterprise-aligned semantic layer.	Define a clear CDS adoption approach, including upfront modelling standards and targeted redesign of high-value datasets to align with certification requirements.

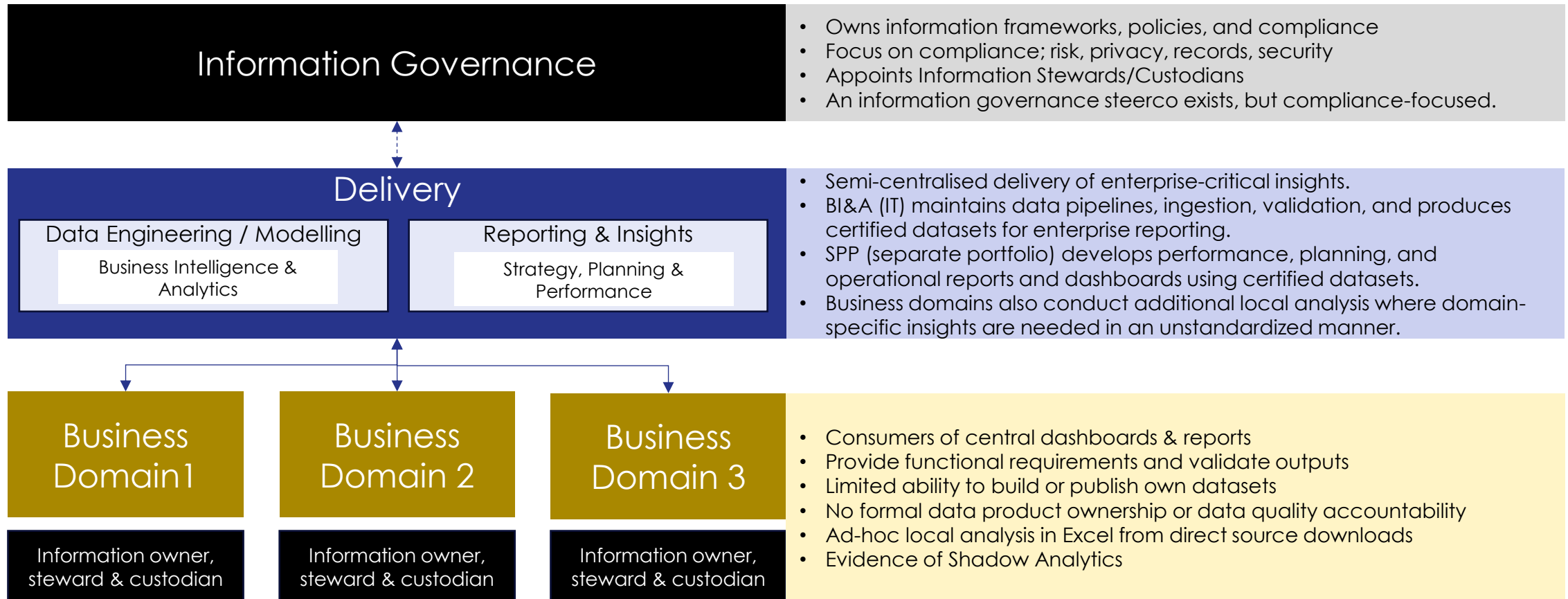
6. Funding & Value Measurement - Finding

While stakeholders recognise the need for uplift, funding is siloed across functions, outcomes are not measured, and there is no enterprise investment model to sustain the foundational capabilities required for reliable onboarding data.

Finding	Detail	Roadmap Consideration
DFV1 Funding & Value Measurement – Ownership and investment in data uplift isn't clear	- Data responsibilities and budgets sit across multiple portfolios (IT, Risk, SPP), making it difficult to fund enterprise-wide capabilities. - UWA lacks a shared target state or data maturity model, so portfolios prioritise different goals.	- Formalise executive ownership for data maturity uplift. - Consider the data strategy and include the definition of a cross-portfolio investment mechanism to fund foundational capabilities needed across the institution. - While centralized funding is most suitable in initial phases, thought will need to go into formalizing a decentralized funding model (i.e. what will the business pay for).
DFV2 Funding & Value Measurement - Data initiatives lack proper project foundations and value frameworks	- Many data initiatives begin without clear scope, ownership, delivery structure, or success measures, making it difficult to track progress or justify continued investment (e.g. the multi-year research data warehouse build). - As a result, multi-year efforts risk drifting in focus, expanding in scope, or continuing without transparent outcomes or value delivered.	Apply consistent project foundations to all data initiatives, including clear scope, ownership, outcomes, and value measures from the outset.

Current Data Operating Model

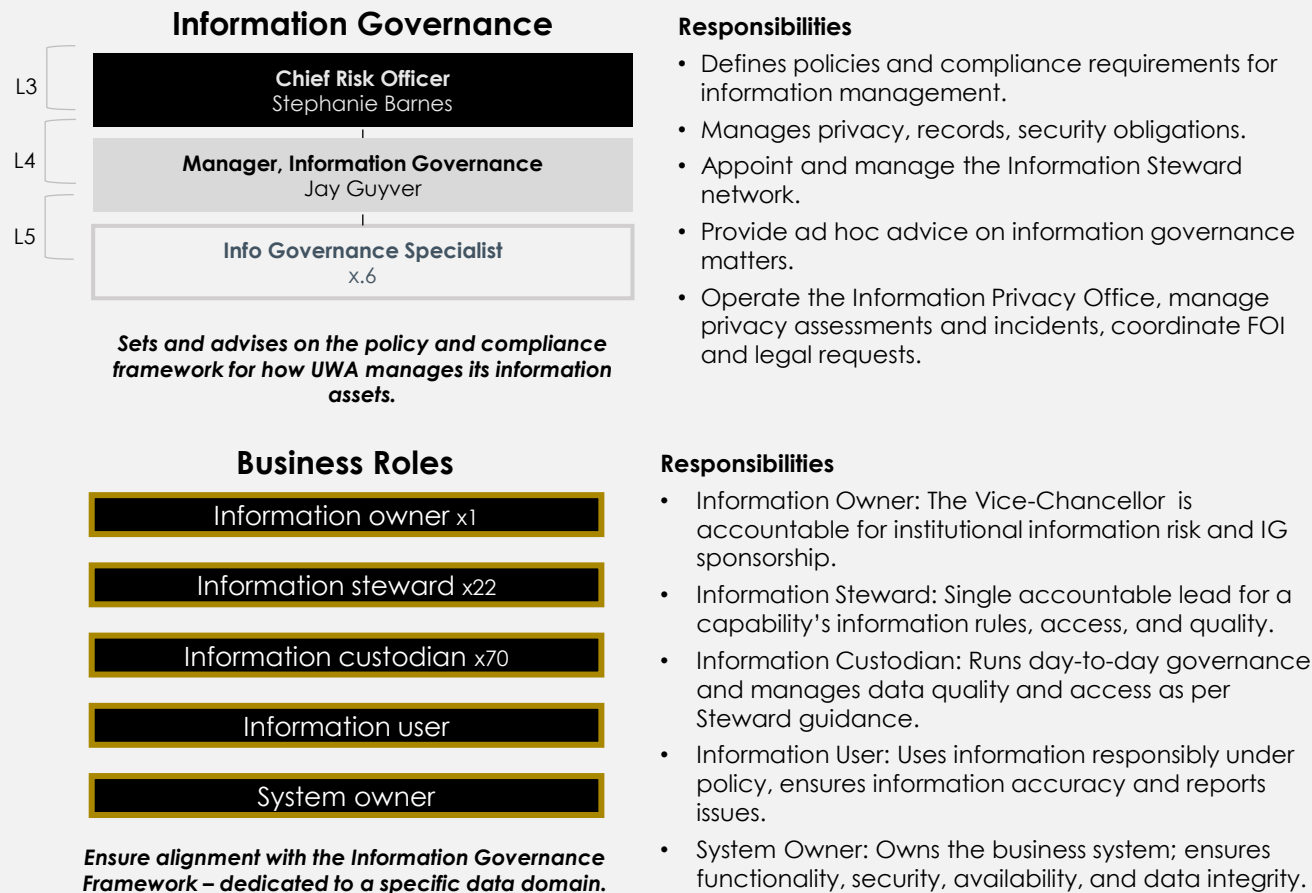
UWA is currently supported by a semi-centralised data operating model, with IT delivering verified data models for a centralised reporting team to utilize to deliver insights to the business.



Information Governance | Size, Structure & Capability

UWA's Information Governance team is a concentrated group that are largely focused on compliance rather than data governance. Its network of business-embedded governance representatives provide a strong foundation to build upon, but the roles aren't well operationalized.

CURRENT STATE STRUCTURE AND RESPONSIBILITIES

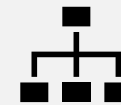


KEY PAIN POINTS / GAPS



Size

- Small team for the breadth of obligations (privacy, FOI, records, security, policy).
- Capacity is largely consumed by compliance work, leaving minimal bandwidth for proactive data governance (definitions, quality, lifecycle).



Structure

- Sits under Risk with a strong compliance lens, not positioned as a partner on data quality and value.
- Blurred boundary between Information Governance and Data Governance – business and IT are unsure who owns rules, definitions, and data quality decisions.
- Information Owners/Stewards/Custodians are nominated but not consistently active, with limited connection to real datasets and processes.



Capability

- Strong at policy, privacy and records, but limited experience, practices and tooling for data governance.
- No structured training, playbooks or forums to equip Stewards/Custodians in their data responsibilities.
- Limited visibility of data risks and recurring issues at a domain or dataset level (no consolidated data risk/DQ view).

Information Governance | Scope

UWA's Information Governance activities are driven by the university's legal obligations and supporting policies. Formal standards around data governance are limited / absent.

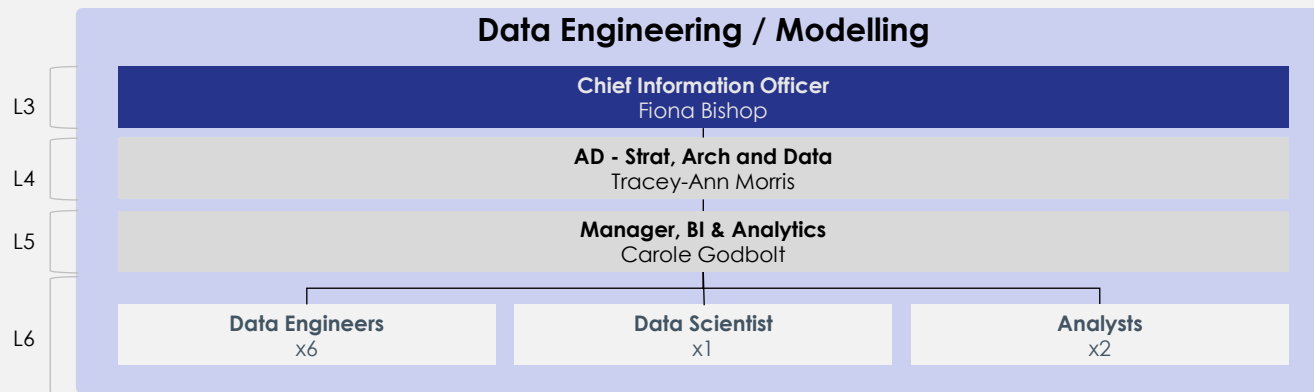
Obligations	UWA Policies & Frameworks*	Implementation Mechanisms
<i>The legislative requirements that govern how UWA must manage, protect, and retain information.</i>	<i>The institutional policies and standards that interpret these obligations into university-wide expectations for information handling.</i>	<i>The processes, roles, and governance mechanisms UWA currently uses to operationalise these policies across the organisation.</i>
Western Australia - Electronic Transactions Act 2011 - Evidence Act 1906 - Financial Management Act 2006 - Freedom of Information Act 1992 - State Records Act 2000 - Privacy and Responsible Information Sharing Act 2024 Commonwealth - Australian Research Council Act 2001 - Higher Education Funding Act 1988 - Higher Education Support Act 2003 - Tertiary Education Quality and Standards Agency Act 2011 - Privacy Act 1988 - Education Services for Overseas Students (ESOS) Act 2000 - Security of Critical Infrastructure Act 2018 - Data Availability and Transparency Act 2022	 Information Governance Framework University-wide rules for creating, managing, using and protecting information assets to ensure alignment with legislation. <i>Info. Gov.</i>  Information Privacy Policy Requirements for collection, processing, retention and disclosure of personal information. <i>Info. Gov.</i>  Information Retention Policy Recordkeeping and authorised disposal requirements across information assets. <i>Records Mgmt</i>  Others Cyber security, responsible use of IT, etc. <i>Records Mgmt, Cyber Sec</i>	1. IG Advisory - Provides on-demand advice on information governance matters. - Maintains a central SharePoint hub with all IG policies, frameworks and guidance. - Uses an IG "toolkit" to ensure new projects meet privacy, retention, and security requirements. 2. Business Embedment - A network of information stewards/custodians support IG compliance, but operationalization of these roles is limited – particularly responsibilities relating to data quality. - The Information Security & Governance Steering Committee (IS&GSC) is where cross-portfolio leaders discuss IG issues & risks. There is no dedicated forum for data governance, so DG has become a default agenda item at IS&GSC. 3. Data Standards & Definitions - A Business Glossary is maintained by SPP to define common business terms and metrics, but its utility for data modelling purposes is limited and it is not tied to overarching data governance standards. - BI & Analytics team sets some rules for Certified Datasets, including naming conventions and transformation logic. Business compliance with these rules is limited due to capability / education gaps. 4. Analytics & Access Controls - The BI & Analytics team manages Power BI workspaces, including naming standards, workspace catalogues, and role-based access rights.

* Note only data-relevant policies have been listed. This is not a comprehensive list of all UWA's policies and frameworks.

BI & Analytics | Size, Structure & Capability

The BI & Analytics team is a relatively small, highly-demanded group responsible for maintaining certified datasets, managing data platforms, and fielding a large volume of ad-hoc requests to support operational and strategic decision-making across portfolios.

CURRENT STATE STRUCTURE AND RESPONSIBILITIES



Manages the enterprise data platform and ensures information is ready for reporting and analysis.

Responsibilities

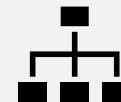
- Design, build and maintain enterprise data pipelines and models, including ingestion from core systems into the Lakehouse and ongoing enhancements.
- Manage and operate the data platform, covering environment administration, vendor management, platform fixes, and performance improvements.
- Ensure data quality, reliability and integrity, including triage, data fixes, validation, and adherence to architecture and modelling patterns.
- Deliver modelling and reporting enablement for new domains and projects (research models, finance models, government reporting, timetabling, etc.).
- Provide BAU support and service operations, including service requests, access management, break/fix, data extracts, release management, and team governance.

KEY PAIN POINTS / GAPS



Size

- The **current team capacity is constrained** with the volume and complexity of ingestion, modelling and service work required across many systems.
- Capacity is stretched across BAU, service requests and major projects, leading to reactive delivery and limited ability to invest in improvements.



Structure

- Fully centralised, so all enterprise data engineering and modelling flows through one team, creating a bottleneck for domains.
- Team members aren't clearly designated to specific work types (project, BAU, service) or domains, resulting in fragmented focus and unclear ownership.



Capability

- **Strong data engineering** skills exist, but limited time to industrialise practices (e.g. automated monitoring).
- **Absence of dedicated architecture and governance roles** forces BI&A to make design and stewardship decisions outside their remit, increasing workload and inconsistency.
- **Capacity constraints limit management and enablement** of domain teams, maintaining dependency on the central team.

BI & Analytics Capacity Deep Dive

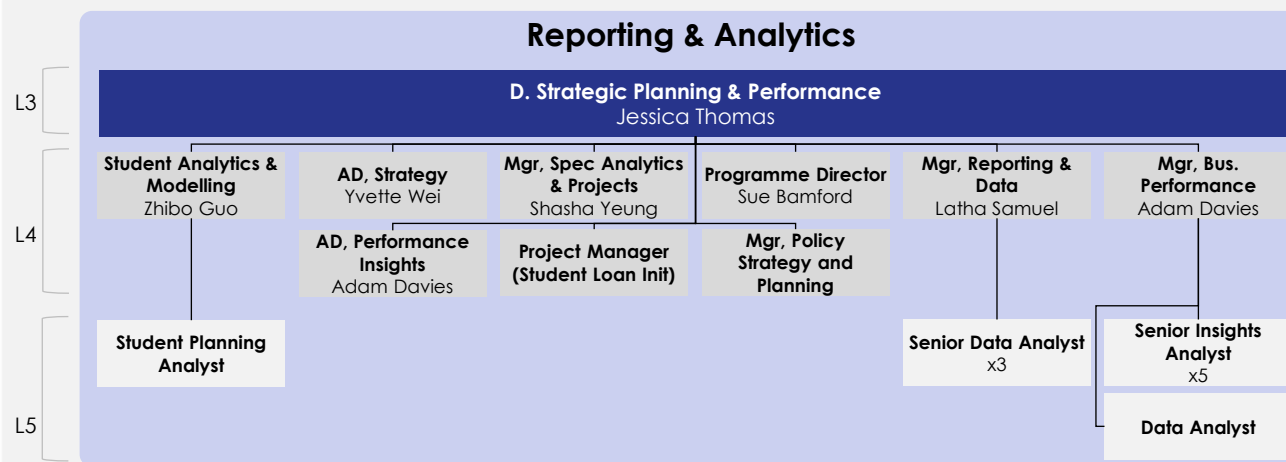
A core issue in UWA's data operating model is the structural overload placed on BI&A. Limited capacity, unclear role boundaries, missing capabilities, and weak foundations all force the team to take on work that should sit elsewhere, leading to reactive delivery and centralised decision-making.

Area	Observation	Description
1. Resourcing	Data Engineering Teams – Roles and Responsibilities	- BI&A's data engineers currently build Gold reporting models for domains, but they often lack the detailed business context needed for accurate logic. - Without a structured feedback loop, business teams recreate or adjust these models locally ("shadow analytics"), leading to inconsistencies and errors that BI&A must later resolve, further slowing delivery.
2. Operations Model	Data Architecture – Roles and Responsibilities not defined.	- Because UWA has no formally appointed Data Architect, architectural decisions (e.g. modelling patterns, semantic structures, and design approaches for new domain) default to BI&A by necessity. - This places the team in the position of making design calls that may fall outside of their scope of expertise and should sit with a dedicated architecture function.
3. Data Governance	Data Governance Role & Responsibilities Gap	- As key data governance responsibilities are not formally owned anywhere these tasks default informally to BI&A. - Business teams rely heavily on BI&A for interpretation and dispute resolution, escalating the team's workload and slowing delivery.
4. Operations Model	BI&A leadership – Roles and Responsibilities	- The BI&A manager currently owns operational triage, modelling oversight, tenancy/access administration, vendor management, governance issues, and team supervision. - This concentration of responsibilities creates a bottleneck, with decisions and priorities often relying on one role, making the team more dependent on central leadership.
5. Change & Training	Training and Enablement Role Gap	- Faculties and administrative units lack a dedicated resource / team to get guidance on dashboards, modelling, or best practice. - Simple issues therefore flow to BI&A, adding to their workload.
6. Operations Model	Demand Management – Project & BAU	- Without a formal intake and prioritisation process, BI&A is frequently asked to support projects without clear resourcing, timelines, or trade-offs. - This results in overcommitment, shifting priorities, and delays to both BAU and strategic work.
7. Resourcing	Capacity Constraints	- BI&A is not currently resourced to deliver its BAU activities as well as the additional responsibilities that fall into their remit given other capability gaps in the business. - The team cannot shift from reactive firefighting to proactive improvement and backlogs form quickly, leading to slow turnaround times.

Reporting & Analytics | Size, Structure & Capability

The Strategic Planning & Performance (SPP) team is UWA's central reporting and insights function, responsible for statutory reporting, enterprise dashboards, and domain-aligned performance analysis that supports planning, compliance and strategic decision-making.

CURRENT STATE STRUCTURE AND RESPONSIBILITIES



Transforms certified data into insights and reports for compliance, performance and decision-making purposes.

Responsibilities

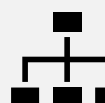
- Deliver statutory and government reporting aligned to census and other compliance cycles.
- Establish strategic KPIs and provide insights to support performance monitoring, budgeting, forecasting and planning across core enterprise domains (e.g. student experience, student analytics, course performance).
- Manage and prioritise ad hoc reporting requests from operational and executive stakeholders.
- Where required, source data through the most appropriate channel for each reporting need (i.e. typically working with IT for EDW updates, but where gaps exist, accessing source-system data or temporary extracts).
- Uphold data governance by applying required security, privacy and access controls (e.g. PowerBI row level access), and engaging Information Stewards to confirm definitions and appropriate data use.

KEY PAIN POINTS / GAPS



Size

- A comfortable analyst cohort overall. Current pressure is driven more by upstream data ingestion / enhancement demand than by reporting capacity.



Structure

- SPP sits in a different portfolio to BI&A (IT), with no formal mechanism to coordinate workloads between reporting and data engineering.
- Spans and layers are misaligned. Several managers have no direct reports, while the Director (L3) has a 9 direct reports, potentially creating an inconsistent leadership load and limiting effective coaching and quality oversight.
- Dependence on datasets from the IT data team creates a bottleneck, as insight delivery is contingent on the capacity of BI&A.

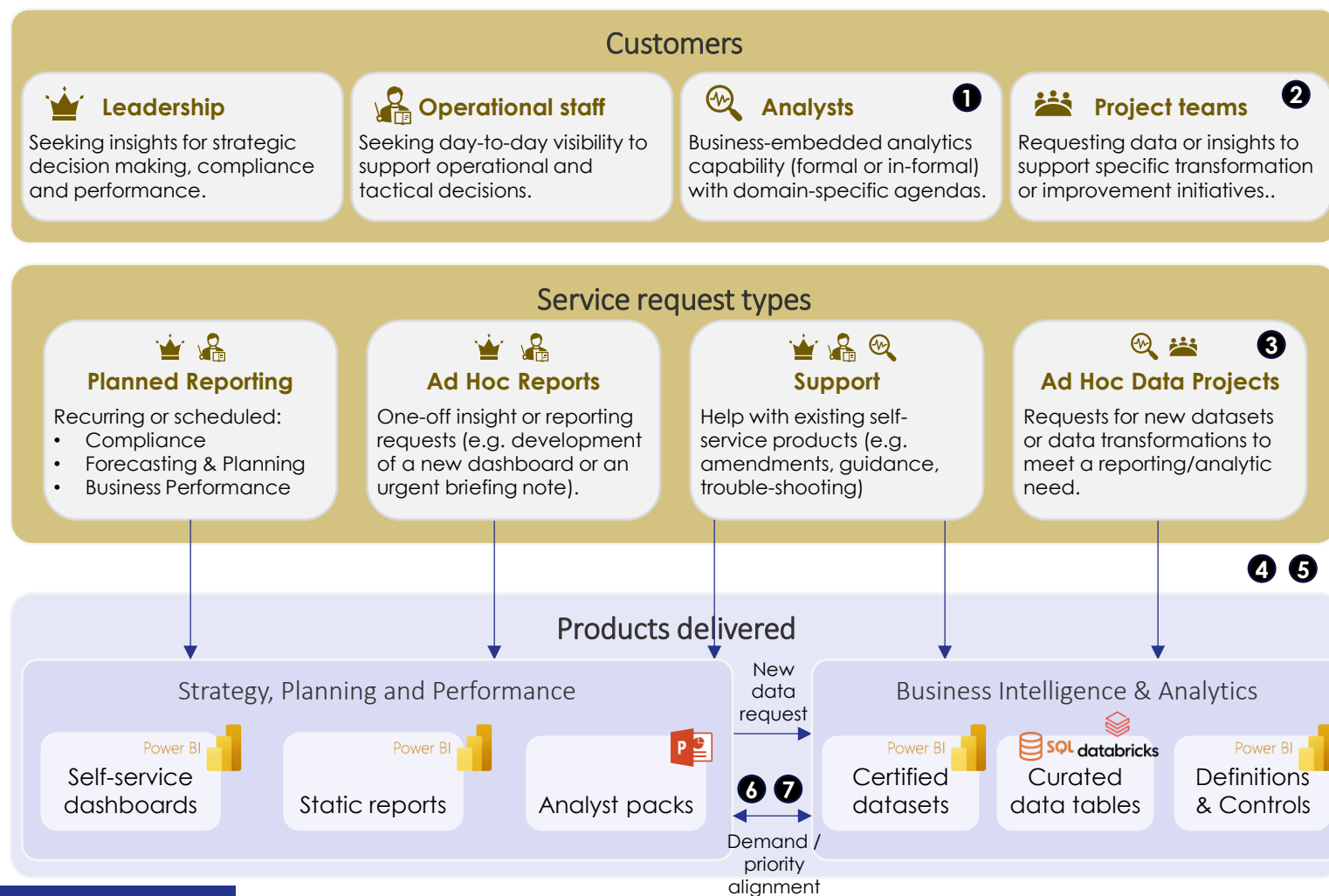


Capability

- Strong PowerBI reporting capability, but limited depth in core data modelling concepts (databricks, star schema design, semantic modelling).
- Gaps in understanding / applying certified dataset guardrails, resulting in duplicated logic and conflicting metrics across dashboards.

Key Process | Demand Management

While SPP & BI&A teams each have relatively established ways of managing their individual work, collaboration between both teams is limited making it hard to see all incoming demand, route requests consistently, and prioritise effort across BAU and projects.

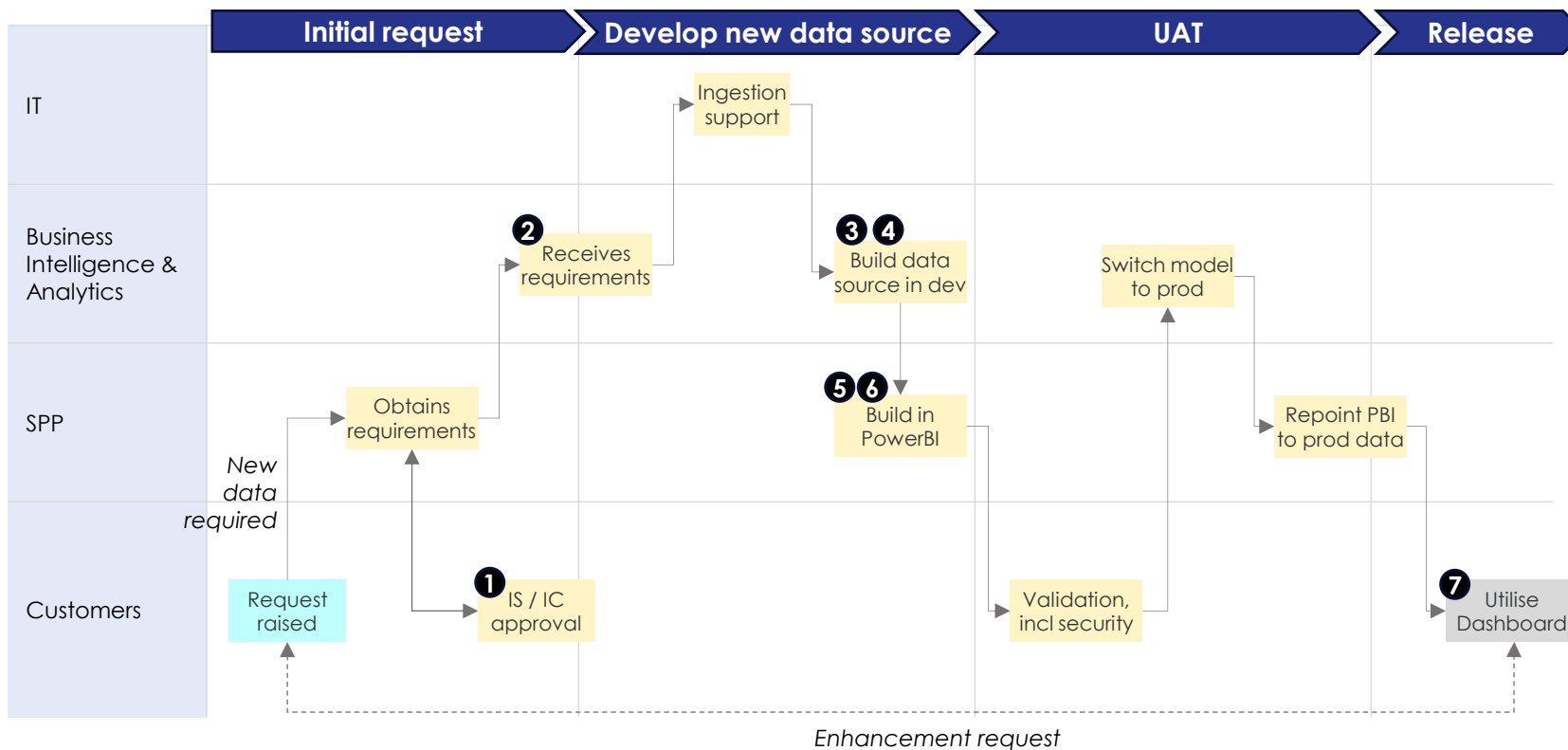


Key pain points / gaps:

1. Low visibility over 'shadow' analysts and their scope or roadmap, making delineation of roles and upcoming demand unclear.
2. IT initiative assessment process doesn't proactively consider downstream data impacts, leading to reactive/urgent requests.
3. BAU engineers in BI&A are frequently pulled into projects without coordinated resourcing, causing BAU bottlenecks and unpredictable availability.
4. Without a clear definition of what the team does/doesn't deliver (i.e. a service catalogue), niche asks enter the intake stream, distorting prioritisation and overloading the team.
5. Requests come through multiple formal/informal channels (ServiceNow, email, Teams, meetings, DMs), making it difficult to manage demand.
6. While BI&A and SPP meet frequently to align on priorities, there is no consolidated roadmap of upcoming requests or expected peaks in demand.
7. There is no clear prioritized, backlog view of work as it is tracked across disparate tools and conversations.

Key Process | Delivery

While SPP and BI&A have a relatively established process to deliver new insights to the business, variations in governance, handover, and quality steps create inconsistencies that make products harder to build, maintain, and improve over time.



Key pain points / gaps:

1. IS/IC roles are inconsistently rolled out, limiting user engagement / ownership.
2. Definitions and business rules aren't centrally governed, resulting in divergent calculations and inconsistent outputs.
3. Upstream schema or ID changes aren't consistently assessed, causing pipelines and models to break mid-build and triggering rework.
4. Because there are no agreed rules for data quality or clear definitions of "done," user testing ends up being the main quality check, leading to rework and delays.
5. There is no consistent mechanism to feed modelling gaps or issues back into BI&A for upstream remediation.
6. Microsoft's platform makes it difficult to retrofit many existing Power BI datasets into a Certified Dataset structure.
7. Platform monitoring is limited, and issues are often identified by users.

** Indicative only*

This process flow only reflects internal requests for new/enhanced insights and does not reflect the diverse requests that need to be met by the SPP & BI&A teams. It has been simplified to demonstrate key pain points along a standard data delivery process.

Process
start

Action

Process
end

2. Develop Phase

Objective

The develop phase aimed to define UWA's target data governance maturity and operating model that their roadmap will aim to deliver on over the next ~5 years.

Activities

Stakeholder Engagement

We have engaged key leadership to define and test the target state operating model and proposed initiatives, including:

- Chief Information Officer
- Associate Director of Strategy, Architecture and Data
- Director of Strategic Planning & Performance

Key Considerations

The target operating model include consideration of:

- UWA's broader business and IT strategic objectives / priorities
- UWA's current data governance maturity & operating model and business literacy
- The diverse data/reporting needs of UWA's domains

Deliverables

- ✓ **Definition of a Target State Operating Model**



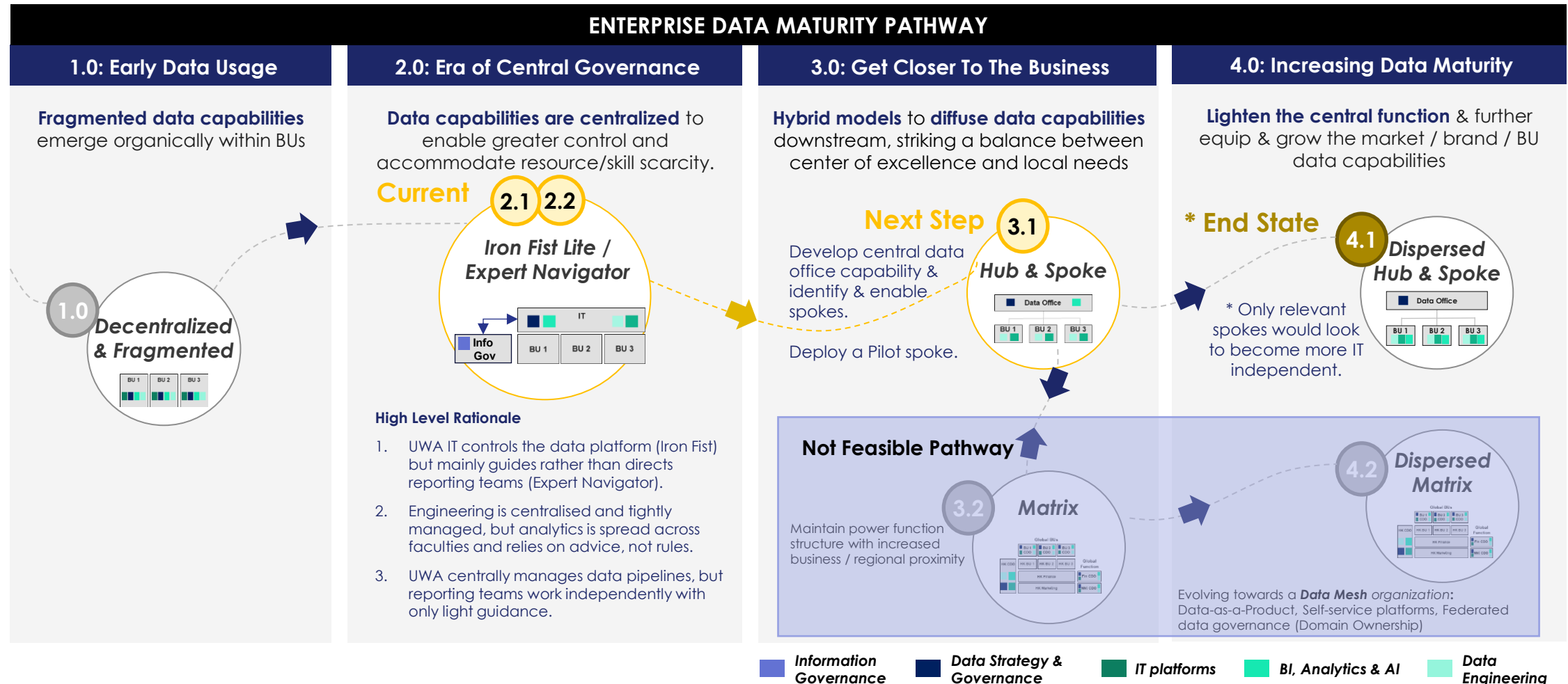
Target State Maturity

As part of the assessment phase, we evaluated UWA across each target dimension and defined the uplift required to reach a fit-for-purpose, future-ready data operating model. The following outlines the key maturity gaps and the corresponding uplift required across each assessment dimension.

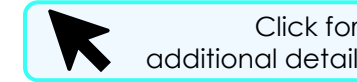
Assessment Criteria	Level 1 - Exploratory	Level 2 - Experimental	Level 3 - Operational	Level 4 - Optimized	Level 5 - Transformed	Uplift Considerations
Strategy & Governance						Establish data strategy, enterprise funding model, prioritisation process, value framework, and investment roadmap.
Business Model						Endorse Data Strategy & Governance activities, activate Owner/Steward roles, formalise decision rights, embed governance forums & HR-backed roles.
Resourcing & Capability						Build domain analytics & governance capacity, appoint stewards, define job families, uplift skills, create training pathways.
Operating Structure						Define Hub & Spoke operating model, including an intake framework, operating rhythms and domain accountability model.
Lifecycle Management						Standardise lifecycle, automate validation, implement change control, reconciliation, shared workflows across systems.
Technology & Tools						Review & implement tooling including - metadata catalogue, lineage, API governance, domain-aligned architecture, reusable pipelines, semantic models.
Data Quality & Metadata						Create DQ framework, define rules, steward ownership, implement DQ tooling, central metadata standards, catalogue rollout.
Sharing & Security						Extend privacy-by-design across domains, standardise access models, implement consistent sharing practices across Spokes.
Culture & Adoption						Enterprise literacy program, domain enablement, governance culture embedding, community of practice, uplift faculty participation.

Target Data Operating Model Architype

UWA's current data operating model is largely centralized, with any decentralized analytics capability largely operating outside of the central rhythm. Given the university's diverse data requirements, the logical next step for the university is to strengthen their central data capability to provide high level oversight and give autonomy to other areas of the business (i.e. 'spokes') to drive their own data needs.



How to Achieve Target State



To enable the Hub-and-Spoke model, UWA must invest in the below three pillars of activity – establishing data governance standards, defining the new operating model, and change management and technology uplift to enable the operating model's successful delivery.



FOUNDATIONS

Define the Rules of the Operating Model

Establish the **data governance framework and associated policies and standards** to ensure all domains structure, govern and interpret data in the same way, including:

- Delineate of data governance vs information governance [Slide 29](#)
- Establishment of enterprise definitions & rules (e.g. glossary, data dictionary, etc) [Slide 30](#)
- Definition of modelling & lifecycle standards outlining how data should progress from Bronze to Gold [Slide 35](#)
- Definition of governance expectations covering metadata, lineage, data quality thresholds and stewardship responsibilities [Slide 30](#)



OPERATING MODEL

Define the Roles, Structure & Interactions Between Teams

Define the Hub and Spoke capability requirements, and how they work together to govern and deliver data and analytics, including:

- Assess and address missing capability within the Hub (i.e. Data Office) [Slides 32-33](#)
- Define priority spoke and introduce spoke-embedded data governance and analytics capability [Slides 34](#)
- Design ways of working to deliver quality and reliable data and analytic [Slides 35](#)
- Define key governance forums that will support the hub and spoke model [Slides 36](#)



ENABLEMENT

Build the Capability to Adopt the Model

Introduce the **tools, platforms, systems, capability and literacy** needed for the target operating model to be successful, including:

- Develop the tools, templates and guidance needed for business-wide compliance with the new standards.
- Strengthen data and analytics platforms, enabling metadata capture, lineage, quality checks and repeatable analytics delivery.
- Introduce change capability to support ongoing adoption of the new practices.
- Build the hub, spoke and wider business skills and literacy required to operate the new model.

Introducing Data Governance

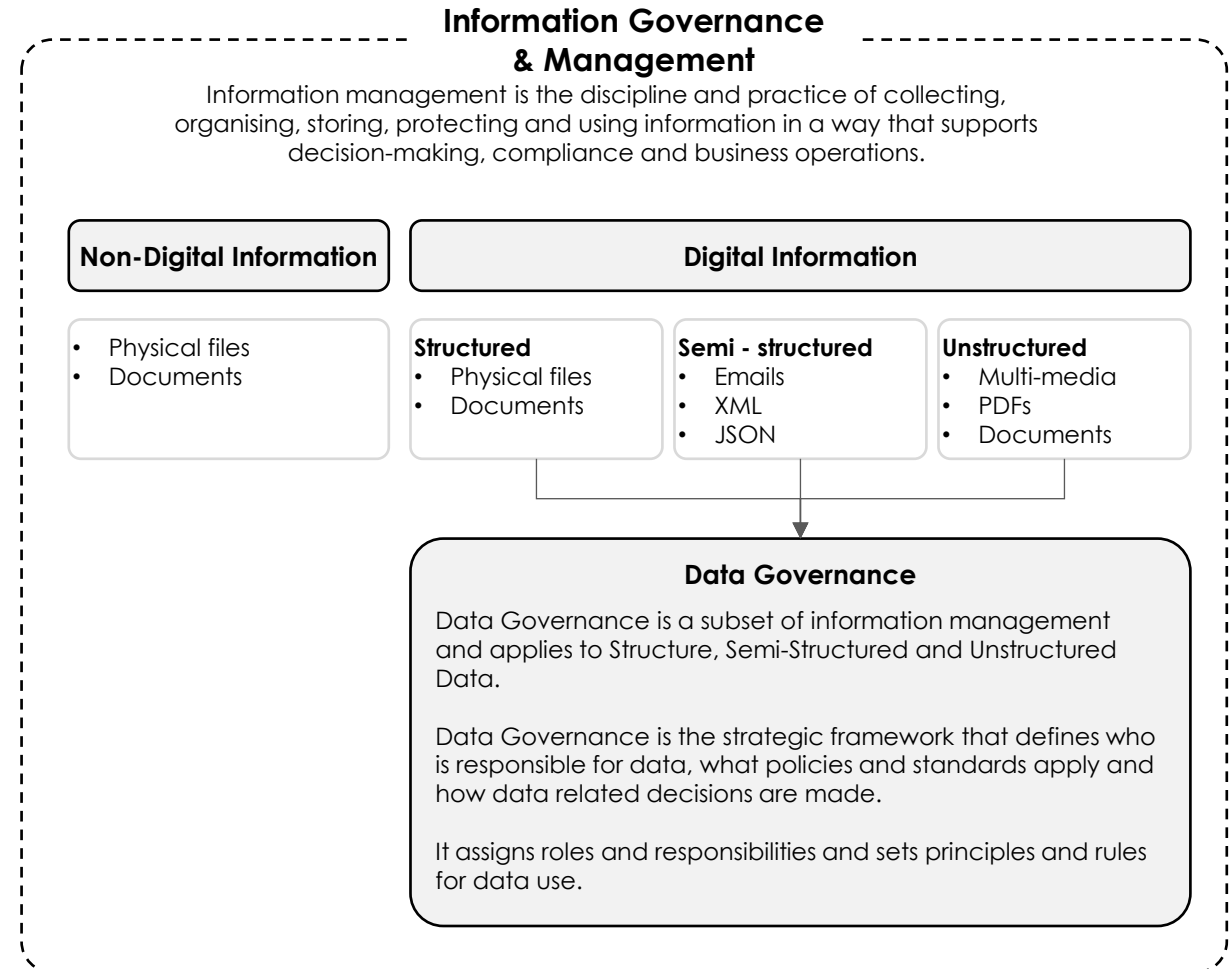
The target state operating model will require a clear definition of UWA's data governance standards and practices, which compliment existing information governance & management policies and frameworks.

In the target state, **Information Governance & Management** remains responsible for governing information across its lifecycle to meet privacy, security, records, and regulatory obligations. Its role is to ensure that:

- Information is protected, retained, and accessed appropriately
- Privacy, security, and records requirements are met
- Compliance and risk obligations are consistently managed

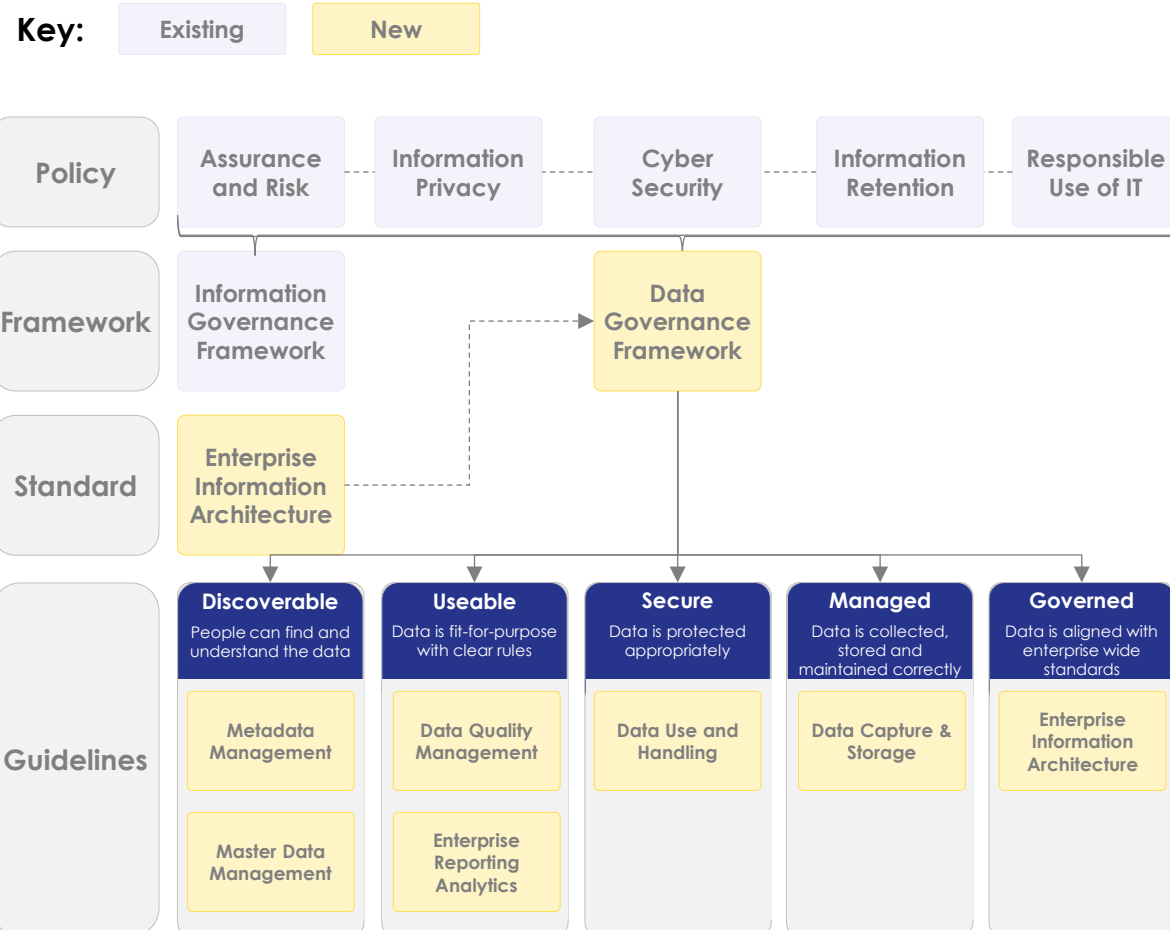
As a subset of Information Governance & Management, **Data Governance** is established as a formal, enterprise capability focused on managing data as an institutional asset. Its role is to ensure that:

- Data ownership and accountability are clearly defined
- Common definitions, standards, and quality expectations are applied
- Data issues are identified and resolved early across the lifecycle
- Data can be confidently reused across domains for reporting, analytics, and AI



Example Data Governance Foundations

Below outlines an indicative view of common, valuable data governance policies, frameworks and guidelines that are generally used to set data governance foundations and how this would sit amongst UWA's existing information governance ecosystem.

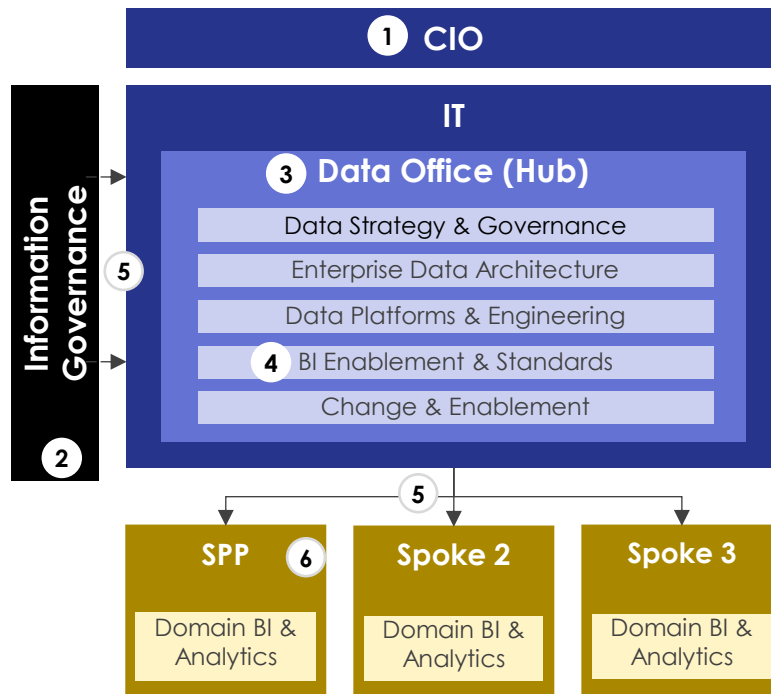


Document	Description	Likely Target Owner*
Data Governance Framework	Defines how UWA governs data, covering roles, decision-rights, ownership, stewardship, lifecycle controls, DQ, metadata, escalations and domain governance.	Head of Data Governance
Enterprise Information Architecture Standard	Sets enterprise structure for UWA's data, including shared models, naming conventions, integration and modelling patterns to ensure consistency.	Enterprise Data Architect
Metadata Management Guideline	Defines required metadata (business terms, lineage, descriptions, owners), documentation standards and expectations for certified datasets.	Data Governance & Quality Specialist
Master Data Management Guideline	Defines rules for managing core entities (student, staff, course, research project), including golden sources, reference data stewardship and change controls.	Enterprise Data Architect
Data Quality Management Guideline	Defines DQ rules, critical data elements, scoring, monitoring, dashboards, triage, and responsibilities for stewards and domains.	Data Governance & Quality Specialist
Enterprise Reporting & Analytics Guideline	Sets reporting and analytics standards, including KPI definitions, semantic model rules, naming conventions, RLS, certification and lifecycle processes.	BI Enablement Lead
Data Use & Handling Guideline	Defines privacy, classification, access, secure sharing, retention and appropriate data use across UWA.	Head of Data Governance
Data Capture & Storage Guideline	Defines standards for data entry, validation, storage, system controls and practices to ensure data is collected and maintained correctly at source.	Data Platform Lead
Enterprise Information Architecture Guideline	Applies enterprise architecture principles to data models, integrations and system changes to ensure alignment with guardrails and interoperability.	Enterprise Data Architect

*Roles defined in subsequent slides.

UWA's Target Hybrid Hub and Spoke Data Operating Model

UWA's target operating model includes a dedicated Data Office within IT that is responsible for developing cross-domain datasets and insights. Domain "spokes" leverage these centrally produced assets to generate their own domain-specific datasets and analytics, supported by central governance and information-governance alignment to ensure compliance.



	Component	Description	Key Responsibilities
1.	CIO	Senior executive owner of data, governance, investment and risk.	- Owns enterprise data & governance strategy. - Provides funding, prioritisation and authority. - Sets enterprise guardrails and escalation pathways. - Provides governance direction and resolves cross-domain issues.
2.	Information Governance	Central function governing information policy, privacy, and compliance.	- Sets mandatory information policies (privacy, FOI, security, retention). - Ensures Hub and domains operate within these rules. - Aligns information governance with data governance. - Provides assurance and escalation for information-related risks.
3.	Data Office (Hub)	Central function for data governance, architecture, engineering, platform operations and BI enablement. No analytics delivery.	- Publish certified, governed enterprise data products for spokes to consume. - Operate the data platform, pipelines, security model, metadata and architecture standards. - Maintain enterprise data governance, quality rules, stewardship frameworks and compliance. - Provide BI enablement (templates, training, RLS patterns, workspace governance) to uplift domain analytics capability.
4.	BI Enablement & Standards	Small enterprise team ensuring BI consistency, governance and standards.	- Maintain enterprise BI standards, patterns, templates and security configurations. - Provide training and support to domain analysts, stewards and custodians. - Govern Power BI workspaces, semantic modelling patterns and certification processes. - Ensure domain analytics adhere to enterprise guardrails for quality, accessibility and compliance.
5.	Governance Committees	Enterprise forums linking all core components.	A range of forums to support: - Executive decision making. - Cross-domain oversight, alignment and escalation of key data risks and issues. - Management of definitions, rules, DQ priorities and enterprise KPIs.
6.	Spoke Teams (Domains)	Domain analytics capability within each spoke / business domain responsible for producing all insights and reporting.	- Build all dashboards, insights, KPIs and reporting products for the domain. - Use Hub-certified data products or build domain datasets aligned to Hub standards. - Own KPI definitions, semantics and business rules, supported by Stewards and Custodians. - Drive decision-making with domain-specific analytics and participate in enterprise governance forums.

Establishing the Data Office ('Hub') Capability (1/2)

The Data Office is the central unit in the target operating model that would provide the platforms, governance standards and practices, datasets and enterprise analytics that enable the Spokes (domains) to operate effectively. It would require the below roles to be allocated/established.

Key Roles	Summary of Responsibilities	Role Rationale	Recommended Transition
Enterprise Data Architecture			
New / Assign Enterprise Data Architect	- Sets data architecture direction, including models, standards, metadata and modelling patterns. - Oversee enterprise reference/master data consistency and ensure alignment across domains. - Provides architectural assurance on new solutions.	No dedicated owner today, but responsibilities are foundational to avoiding divergent models in a hub & spoke model by creating a unified Lakehouse blueprint with clear standards.	- Assign role and augment to existing team or potential new hire. - Require definition of modelling patterns, lineage and metadata governance foundations in Phase 1. - Role becomes the "control point" for domain modelling to ensure consistency with future ERP changes.
Data Governance & Strategy			
New Head of Data Governance	- Own and operationalise governance policies, standards, definitions and decision forums. - Lead university-wide network of Stewards/Custodians and manage governance escalations.	No dedicated owner today so data governance is reactive and political without a central authority. A central role mitigates current culture issues, reduces key person dependency, and provides structure.	- Introduce a new resource to establish as single data governance authority across UWA and support the enablement of the foundational governance items. - Shift all governance responsibilities out of BI&A/SPP into the Hub. - Re-purpose the existing Information Steward/Custodian network to include data governance activities.
New Data Governance & Quality Specialist	- Maintain enterprise quality rules, DQ dashboards, and support remediation workflows. - Partner with Stewards/Custodians on quality monitoring, issue triage and rule enforcement.	Time-consuming data governance maintenance activities can be taken from data engineers and analysts, giving them space to focus on their roles.	- No immediate additional hiring. A new role can be introduced once the Head of Data Governance has established the foundations (e.g. DQ tooling). - Specialist will support the definition of critical data elements (CDEs): business glossary, dictionary, quality rules.
Data Platforms & Engineering			
Assign Data Platform Lead	- Operate and optimise the data platform including performance, RLS, and security. - Manage engineering standards, platform hygiene, CI/CD and monitoring.	Today's platform ownership is unclear. A dedicated platform manager ensures velocity, consistency and removes dependency on analysts building pipelines.	- Assign role to a member of the IT engineering team OR hire if not available (role cannot be shared with BI Enablement Lead). - Introduce platform Service Level Objectives, incident response, and security baselines. - Engage with Hubs through forums to ensure demand and delivery alignment.
Existing Data Engineers	- Build & maintain ingestion pipelines (Bronze/Silver/Gold) and orchestrate data movement. - Troubleshoot ingestion, maintain transformations and support migrations to Lakehouse. - Support some internal IT related reporting and analytics requirements (Internal IT Operations only)	Engineering capability is currently highly overloaded with a wide-range of responsibilities that should not fall within their scope of work. More focus on delivery rather than governance and co-ordination.	- Shift existing data engineers (~6) to focus on ingestion of Bronze, Silver & Gold data sets. - No new FTE required if focusing engineers strictly on ingestion + platform enablement. - Formalize engineering standards: prioritization, testing, doc, code review, DevOps, version control.

Establishing the Data Office ('Hub') Capability (2/2)

The Data Office is the central unit in the target operating model that would provide the platforms, governance standards and practices, datasets and enterprise analytics that enable the Spokes (domains) to operate effectively. It would require the below roles to be allocated/established.

Key Roles	Summary of Responsibilities	Role Rationale	Recommended Transition
BI Enablement & Standards			
New / Assign BI Enablement Lead	- Lead the BI Enablement & Standards team, ensuring enterprise-wide BI delivery quality and consistency. - Maintain enterprise BI standards, templates, semantic modelling patterns, RLS/security rules and workspace governance. - Provide expert guidance, coaching and uplift to spokes and developers building dashboards across domains and ensure central alignment. - Ensure enterprise-certified data products and semantic models are used correctly and consistently by all spokes.	- Hub-and-spoke models require strong central guardrails for BI consistency, quality and compliance — without delivering analytics. - This role ensures domains build analytics safely, correctly and consistently, reducing duplication and risk.	- Assess if existing resources can fulfil this role. - Focus the role on enterprise standards, security patterns, templates and BI governance across all domains. - Establish structured engagement with domain analytics teams (e.g., SPP, Planning, Finance, HR) to support maturity uplift.
Assign BI Standards & Modelling Specialists	- Maintain and evolve enterprise BI standards, including modelling patterns, DAX standards, naming conventions, workspace governance and RLS frameworks. - Provide practical enablement for spokes (training, code reviews, model reviews, best practice guidance). - Ensure hub-certified data sets, semantic models and metadata are correctly consumed by domain teams. - Perform quality checks and compliance assessments to ensure BI outputs across domains meet enterprise guardrails.	- As analytics delivery moves to spokes, UWA requires specialist capability centrally to ensure BI consistency, reduce risk, avoid fragmentation, and uplift domain maturity. - These specialists replace the need for central report developers by enabling every domain to build high-quality analytics.	- Maintain 1–2 FTE focused on BI standards, workspace governance, and enablement (not report development). - Define and publish BI enterprise standards: semantic layer rules, definitions, naming, modelling templates, RLS patterns. - Position the BI Enablement Team as a Centre of Enablement, not a bottleneck or reporting function.
Change & Enablement			
New Data Literacy Lead	- Develop literacy playbooks, training, onboarding materials and self-service guides. - Drive maturity uplift across Stewards, Custodians, Analysts, and Spoke teams.	Data literacy is currently ad hoc and significantly dependent on personal coaching from a few individuals. A dedicated role will be required to set the training / literacy agenda and expectations as spokes are rolled out.	- Assign new hire or use internal training resources. - Deliver a structured program: literacy level expectations, persona-specific learning paths, domain training curriculum.
New Training & Adoption Specialist	- Deliver training, onboarding and support for governance processes and analytics use. - Support change management for platform and model updates.	As spokes are rolled out, a dedicated training delivery role is necessary to support the Data Literacy Lead and reduce dependency on central teams.	- Initially combine with Literacy Lead (same individual) until scale increases. - Establish standard adoption packs for each new Spoke domain.

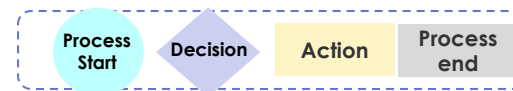
Spokes Overview

Spokes provide domain expertise, validate definitions, manage local data quality, build operational reporting, and ensure governance is applied at the business-unit level, while the Data Office Hub provides the enterprise foundations and standards. A Spoke exists where a domain owns key processes, generates important data, needs local insights, and has enough capability to participate in governance and work with the Data Office Hub.

Spoke Role	Description	Core Responsibilities
Information Owner (Executive in the Domain)	The senior leader accountable for the domain's information and outcomes (e.g., Director of Student Admin, CFO, HR Director, Director of Research).	• Approve domain KPIs, rules and priorities • Ensure compliance with governance standards • Champion adoption in their domain • Escalate issues to the Data Governance Board
Information Steward	Senior functional specialist with deep domain knowledge (e.g., Student Records Lead, Research Grants Lead, HR Operations Lead).	• Own business definitions & domain rules • Validate data quality requirements • Approve changes to KPIs, definitions, business processes • Represent the domain in governance forums • Ensure staff follow governance and quality practices
Information Custodian	Operational staff who manage data in systems (Callista, HRIS, Finance, CRM, LMS).	• Ensure correct data entry and process integrity • Monitor local data quality issues • Apply retention, privacy, access and classification rules • Log DQ issues and support remediation • Approve local access requests (within governance rules)
Spoke / Domain Analytics & Reporting, Data Analysts	Analytics capability embedded in the domain (may be 1–2 people). Responsible for all domain insights, dashboards and reporting.	• Build all domain dashboards, KPIs, insights and operational reporting using Hub-certified data products. • Apply domain-specific logic, rules and business context to produce meaningful insights. • Ensure KPI alignment and semantic consistency by working with Stewards and Custodians. • Identify data quality issues, raise them through governance channels, and support issue resolution.
Power User / Super User	Highly knowledgeable user of systems or reporting tools; first point of contact for colleagues.	• Support local staff with data entry and reporting questions • Reinforce data standards and good practices • Help resolve minor local issues before escalation • Provide local training and awareness
Data Consumers	Staff who consume and enter data day-to-day (academics, advisors, administrators).	• Enter accurate data in systems • Use dashboards appropriately • Follow governance rules • Report issues or discrepancies early

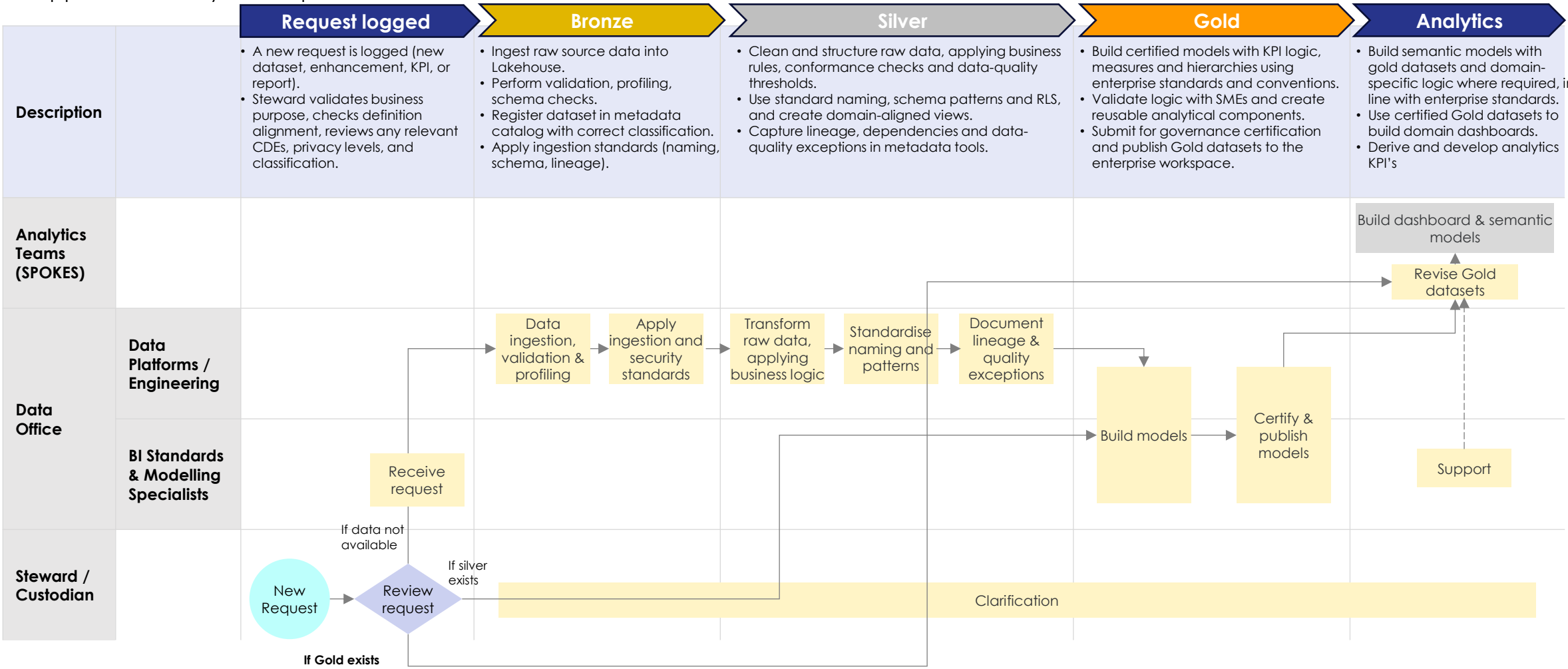
A domain should be a Spoke if it meets majority of the following:

- 1 Owns Critical Business Processes**
(e.g., Student Lifecycle, Research, HR, Finance, Faculty operations)
- 2 Generates or Manages High-Value Data**
Data that impacts reporting, funding, compliance, or cross-domain KPIs.
- 3 Has Domain Expertise Not Held Centrally**
Specialist knowledge required to validate definitions, rules, quality, and context.
- 4 Requires Local Operational Reporting**
Regular day-to-day dashboards or reports that central teams cannot practically produce.
- 5 Has Capacity for at Least Minimal Analytics or Stewardship**
At least 1–2 people who can act as Steward, Custodian, or Analyst.
- 6 Needs to Participate in Governance**
Input into definitions, business rules, quality requirements, and data decisions.



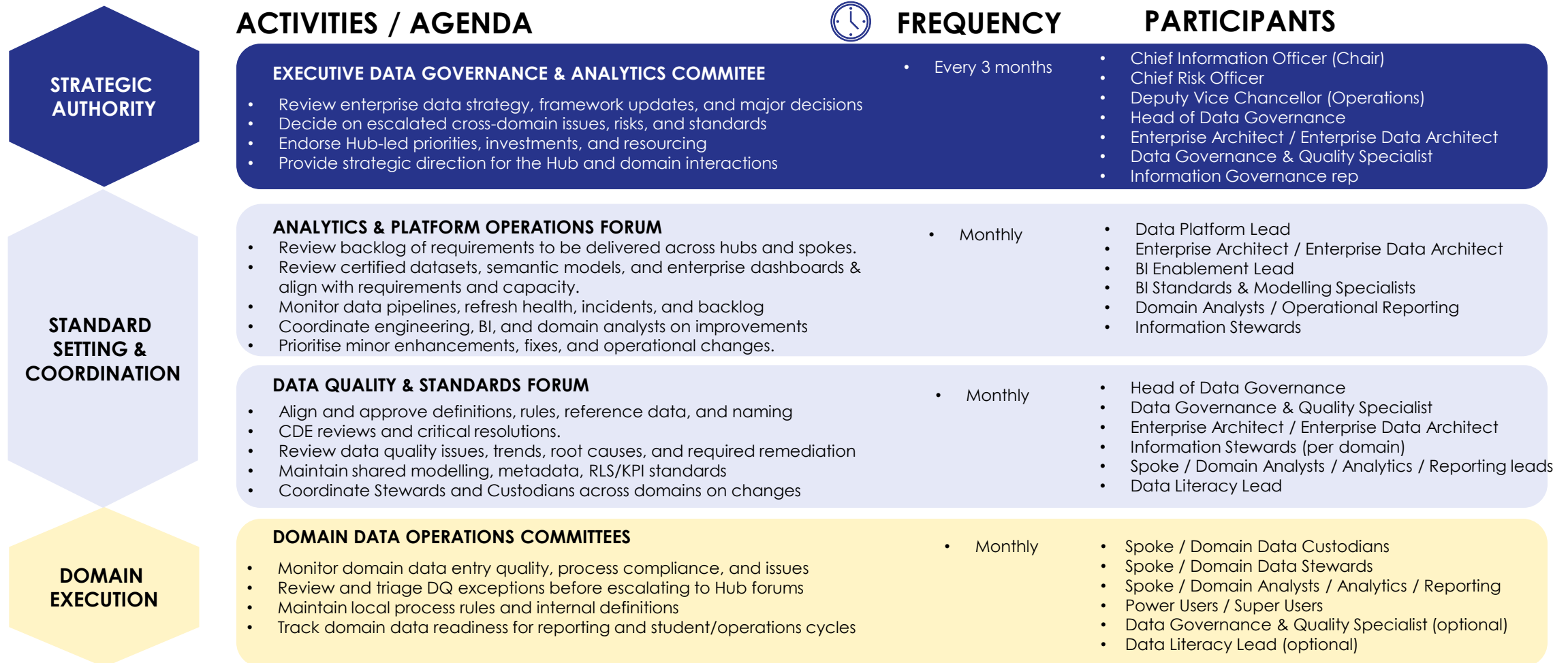
Responsibilities Across the Bronze–Silver–Gold Lifecycle

In the target operating model, the Data Office (Hub) engineers are responsible for the preparation and operation of Bronze, Silver, and enterprise Gold data layers. The BI Enablement team certifies enterprise Gold datasets and connects them to spoke teams. Domain teams then extend these certified Gold assets through domain-specific semantic modelling and KPIs, within enterprise standards and with BI Enablement guidance, to support their analytical requirements.



Example Data Governance Forums

Dedicated governance forums will need to be established to manage and maintain compliance with UWA's new data governance standards. This will likely require adjustments to existing cadences to avoid overlap (e.g. the Information Security and Governance Steering Committee will no longer need to discuss data quality issues or BI&A backlog).



3.Roadmap

Objective

The following section provides a roadmap of activities to address UWA's more critical pain points and establish the foundations needed to transition to their target state maturity and operating model.

Activities

Stakeholder Engagement

We have engaged key leadership to validate the roadmap, including:

- Chief Information Officer
- Associate Director of Strategy, Architecture and Data

Key Considerations

The proposed roadmap include consideration of:

- UWA's current data governance maturity & operating model and business literacy
- Critical pain points, and associated risk levels and/or impact.
- Current or planned IT/business initiatives

Deliverables

- ✓ **Data Op Model Roadmap (~2-5yrs)**
- ✓ **One Page Plans for Key Immediate Priorities**



Roadmap | Introduction

The roadmap outlines how UWA can strengthen the maturity of its data governance and shift towards a hub-and-spoke data operating model. The roadmap reflects a staged path – initially focused building the ‘hub’ with the appropriate standards, processes, roles and skills. From there, each spoke will be introduced in a staged manner to minimize change impacts and ensure the operating model supports each domain's unique needs.

The foundational activities in this roadmap will inform UWA's Data Strategy and guide allocated investment into uplifting the university's data maturity.

KEY ACTIVITIES

Data Governance 	Operating Model 	Resourcing & Capability 	Culture & Adoption 	Systems & Technology 
Build the core governance foundations (e.g. roles, standards, and tooling) that make data ownership and decision-making consistent across UWA.	Design and establish a Hub-and-Spoke model that formalises the Data Office and introduces coordinated ways of working across domains.	Right-size / uplift data capability with the modelling, stewardship, and analytics skills needed to operate the new model effectively.	Drive shared accountability by providing training, playbooks, and communities of practice that help staff understand and apply the new standards.	Activate the platform capabilities (e.g. metadata, lineage, quality, and semantic models) that underpin reusable data and AI.

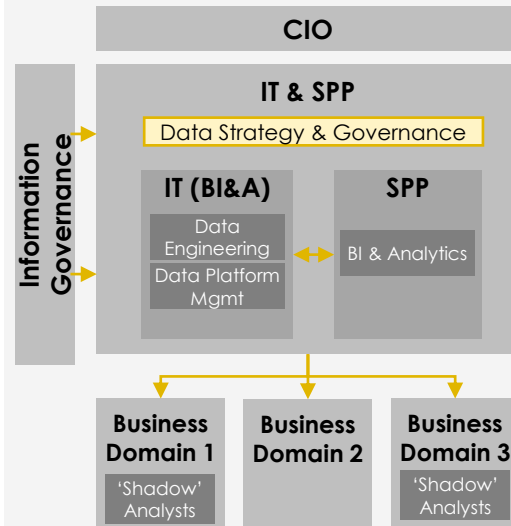
HORIZONS

	P1 - Data Foundations Year 1	P2 – Maturing Domains 1-2 Years	P3 - Innovation 2+ Years
OBJECTIVES	Put in place the governance, architecture, operating model and core capabilities needed to enable a scalable hub-and-spoke model.	Test and refine the hub-and-spoke model by maturing existing domains.	Mature the federated model, expand spoke adoption and evolve advanced data capabilities and platform automation.
FEATURES	Governance framework and roles; foundational policies and standards; hub-and-spoke blueprint; Data Office stand-up; modelling and semantic layer foundations; metadata and lineage tooling; literacy and change uplift.	Expanded definitions, DQ operations, metadata tooling, full Data Office, first Spoke, semantic layer build, certified datasets, analytics optimisation, AI pilots, Copilot rollout, training programs	Federated governance; multiple spokes operating within guardrails; enterprise data products and feature stores; automated quality and lineage; scalable pipelines; continuous improvement.

Roadmap Transitional phases

Immediate

Phase 1.1 – Data Foundations Planning & Commencement

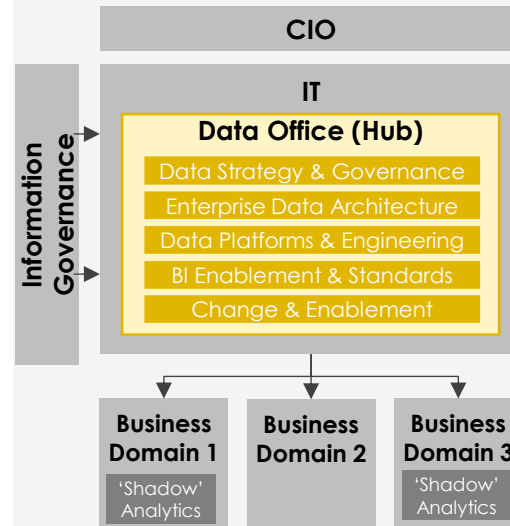


Review and define foundations:

- Data strategy review & alignment
- Establish core data governance foundations (i.e. policies & standards)
- Define the data operating model, including key forums.
- Uplift existing data demand management processes.
- Assess & define data governance tools and platform requirements.
- Identify and define critical data elements (CDEs)
- Plan the IT data team's uplift & re-brand.

0-1 Year

Phase 1.2 – Data Foundations Enablement

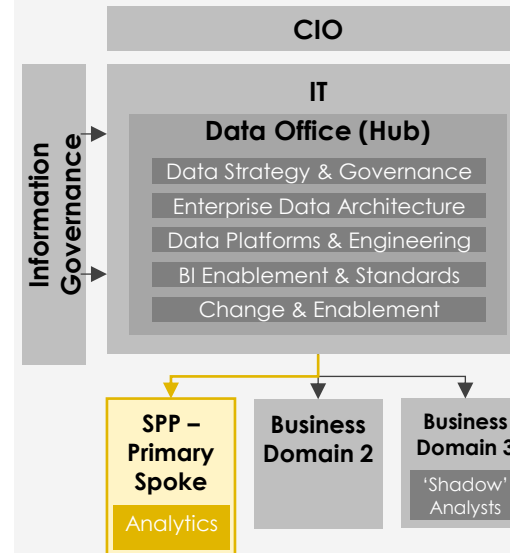


Formalise governance and embed supporting structures:

- Embed data governance frameworks and policies into workflows and tools.
- Introduce and enable Data Office roles with training and change support.
- Implement domain governance roles with supporting processes and tools.
- Re-brand and uplift the IT data team as the Data Office.
- Communicate key changes to the broader business.

Year 1-2

Phase 2 – Maturing Domains

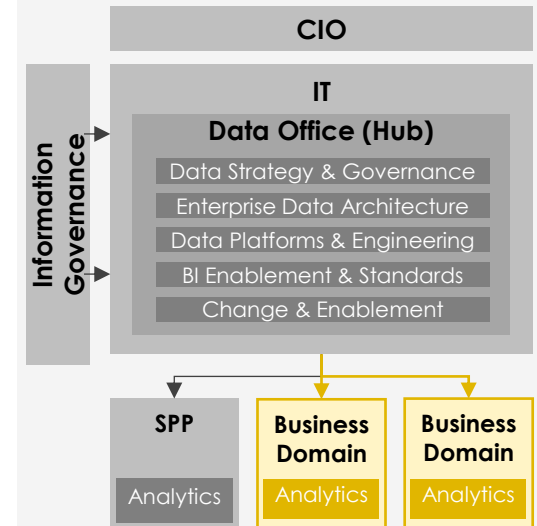


Pilot the hub-spoke model with one domain:

- Define detailed workflows, roles, and governance interactions.
- Uplift domain capability to operate within the model.
- Configure the spoke's technical environment.
- Trial new ways of working and refine the spoke pattern for scale.

2+ years

Phase 3 – Innovation



Gradually expand hybrid federated models to other domains:

- Expand the hub-and-spoke model to additional domains.
- Implement future-state architecture and guardrails needed for a federated model.
- Automate governance, quality, and lineage controls.
- Embed continuous improvement across spokes and the Hub.

Phase 1.1 – Data Foundations Planning

Low	Medium	High	None
●	●	●	-

Phase 1.1 focuses on establishing the essential foundations required for UWA to manage, govern, and scale its data effectively. This phase sets the baseline for all future analytics, digital, and AI activity by defining the data strategy, strengthening governance, clarifying roles, and creating the initial operating model. It ensures UWA has consistent standards, trusted definitions, clear ownership, and a capable analysts - enabling reliable enterprise data, improved decision-making, and a structured pathway toward a hybrid hub-and-spoke model.

Ref	Area	Title	Description	Comment	Effort	Complexity	External
WP_1	Data Governance	Data Strategy Review & Development	Assess UWA's current data landscape and define a forward-looking data strategy that sets the vision, priorities, target architecture and investment pathway for governance, analytics, AI readiness, and operating model maturity. Establish alignment with IT strategy and institutional objectives and identify the foundational capabilities required to support scalable data and AI initiatives.		●	●	
WP_2	Data Governance	Data Governance Framework, Guidelines & Policies	Develop and formalise the university-wide Data Governance Framework, including roles, responsibilities, standards, definitions, privacy-by-design principles, data quality rules, and lifecycle management. Develop or update supporting guidelines and policies to provide clear, consistent governance guardrails across domains.	Recommended foundations on slide 31. Full page initiative provided.	●	●	✓
WP_3	Operating Model	Data Operating Model Design	Define the target enterprise hub and spoke data operating model, detailing how the central hub, business domains, and governance roles work together. Establish key processes, responsibilities, and engagement pathways to enable consistent data delivery, governance, and scalable analytics and AI.	High level design in 'Develop phase' section. Full page initiative provided.	●	●	✓
WP_4	Operating Model	Define & Establish Data Governance Forums	Define and launch the core governance structures required to manage data effectively, such as a Data Governance Board and Standards & Quality Forums. Set clear mandates, decision rights, meeting rhythms, escalation paths, and alignment to the Information Governance model.	Example forums on slide 37.	●	●	
WP_5	Data Governance	Critical Data Element Identification	Identify and prioritise the university's most critical data elements across key domains (e.g., Student, Research, HR, Finance). Document definitions, business rules, lineage, ownership, quality expectations (aligned with guidelines), and usage patterns to establish authoritative enterprise data and support analytics and AI accuracy.		●	●	✓
WP_6	Tooling	Review Data Governance Tools	Evaluate existing tools and repositories (e.g., purview, glossary, definitions catalogues, lineage trackers, metadata stores) for suitability. Determine gaps and select appropriate technologies to enable metadata management, lineage visibility, data quality monitoring, certification workflows, and scalable governance automation.		●	●	
WP_7	Operating Model	Data/Analytics Demand Management Uplift	Improve SPP and BI&A demand management by introducing a single intake process, shared prioritisation criteria, and a visible, jointly-owned backlog, supported by a weekly delivery cadence to review priorities, dependencies, risks and resourcing. This will stabilise data service delivery today, while creating the structure needed to smoothly transition into the future hub-and-spoke operating model.		●	●	

Phase 1.2 – Data Foundations Enablement

Low	Medium	High	None
●	●	●	-

Phase 1.2 focuses on operationalising UWA's data foundations by embedding governance, enabling key roles, deploying quality tooling, and launching the Data Office. This phase shifts the model from design to practice - activating stewards and custodians, standing up governance forums, and aligning the EDW migration to new standards and processes. The outcome is consistent execution, clearer accountability, and a scalable operating model ready to support future spokes and broader analytics and AI uplift.

Ref	Area	Title	Description	Comments	Effort	Complexity	External
WP_8	Systems & Technology	Deploy Data Governance Tools	Address tooling gaps identified in WP_6 to enable data governance practices. This includes introducing and/or configuring systems to support metadata & glossary, lineage, data quality and semantic modelling & certification capabilities. This will be an ongoing activity that will start by supporting data office activities and evolve to support domains as they are onboarded to the new operating model.		●	●	
WP_9	Resourcing & Capability	Establish Data Office	Hire or assign the roles required to stand up the Data Office, including architects, engineers and governance roles. Further define responsibilities, decision rights, and operating rhythms where required.		●	●	
WP_10	Culture & Adoption	Hub Training	Deliver training for all Hub roles on governance standards, modelling practices, workflows, quality rules, and supporting tools to ensure consistent execution across the central team.		●	●	
WP_11	Resourcing & Capability	Implement Domain Governance Roles	Activate Steward, Custodian, and business System Owner roles by formalising decision rights, handoffs, and responsibilities.		●	●	
WP_12	Culture & Adoption	Domain Foundations Training	Provide foundational training to domain Stewards, Custodians and System Owners on their responsibilities for data quality, definitions, and governance.		●	●	
WP_13	Operating Model	Embed Data Governance in Operational Workflows	Introduce defined governance rules into day-to-day Data Office and Domain workflows. This includes embedding standards (definitions, quality rules, access, lifecycle, lineage) into data practices like analytics, data engineering, reporting, and data entry. Ensure governance rules are followed in how models are built, how data is entered and published, how definitions are approved, and how issues are resolved.		●	●	
WP_14	Data Governance	Deploy CDE governance	Roll out targeted CDE data quality monitoring, data dictionary, business glossary, and lineage tools to provide transparency of data health and provenance. Establish rules, dashboards, and alerts that support proactive quality management and feed into governance forums and stewardship processes.		●	●	
WP_15	Systems & Technology	Review & support EDW Migration Project	Review the existing EDW migration program to assess its impacts, risks, and dependencies on the new data operating model, ensuring BAU continuity and understanding implications for current teams and governance structures.	Full page initiative.	●	●	✓
WP_16	Culture & Adoption	Data team re-brand	Introduce the Data Office re-brand through a university-wide change and communications program that explains their roles, how staff should engage with them, and who to contact in their domain for data issues, ensuring all units understand how to operate within the new data ecosystem.		●	●	

Phase 2 – Maturing Domains

Low	Medium	High	None
●	●	●	-

Phase 2 focuses on rolling-out the hub and spoke operating model with a high priority and mature business domain. This phase will focus on establishing domain-level rules, tools and workflows that align with data governance standards set in earlier phases. The interactions between the Data Office (hub) and the pilot domain will be designed in greater detail during this phase, tailored to the domains needs and context.

Ref	Area	Title	Description	Effort	Complexity	External
WP_17	Operating Model	Spoke Pilot Identification	Prioritise business domains for onboarding into the hub-and-spoke model based on data maturity, strategic value, and readiness to adopt shared standards. SPP and Load Planning are the natural first spokes, as they already operate closely with the Hub to deliver enterprise insights. This initiative will focus on formalising their roles, responsibilities, and alignment to the operating model before expanding to other domains.	●	●	
WP_18	Operating Model	Detail Spoke Operating Model	Apply the enterprise hub-and-spoke operating model to the selected pilot domain by designing its detailed day-to-day interactions with the Data Office (Hub). Establish clear hand-offs between stewards, custodians, analysts and Hub governance roles. Use early interactions to test, refine, and streamline the operating model so that it becomes a repeatable, scalable pattern for future spokes.	●	●	✓
WP_19	Resourcing & Capability	Spoke Data Capability Uplift	Assess the pilot spoke's readiness (i.e. skills) and existing analytics demand / priorities. Identify key skills/roles needed to successfully operate within the hub-and-spoke model (e.g. local analysts or engineers, modelling training, etc). Roll out the required uplift activities to ensure the spoke has the skills, support, and capacity to function effectively during the pilot.	●	●	
WP_20	Systems & Technology	Configure Spoke Tooling	Set up the pilot spoke's technical environment so it can work effectively within the hub-and-spoke model. This includes configuring its workspace, access and security roles, lineage and data quality views, and the use of certified semantic models. While tailored to the spoke's needs, ensure set up is aligned with enterprise standards.	●	●	
WP_21	Culture & Adoption	Spoke Data Literacy Uplift	Deliver communications that clarify the hub-and-spoke model, including roles, responsibilities and how the spokes engage with the hub. Strengthen data literacy across domains and promote clear, simple mechanisms for raising data issues or risks.	●	●	
WP_22	Operating Model	Spoke Roll-out	Roll-out the pilot spoke by putting the designed ways of working into practice. Observe how these behaviours work in practice, identify where adjustments are needed, and refine the operating model based on what is learned.	●	●	

Phase 3 – Innovation

Low	Medium	High	None
●	●	●	-

Phase 3 expands the hub-and-spoke model to additional domains and introduces more advanced capabilities to strengthen governance, automation, and scalability. This phase focuses on maturing the platform architecture, improving data quality and lineage processes, and enabling domains to operate with greater autonomy while maintaining enterprise guardrails.

Ref		Title	Description	Effort	Complexity	External
WP_23	Operating Model	Scale Hub-and-Spoke to Additional Domains	Identify where shadow analytics currently exist and which areas have the strongest data needs, then roll out the refined hub-and-spoke model to new domains in priority order. Tailor the model to each domain's maturity and provide targeted training in modelling, governance and data literacy to support successful adoption.	●	●	
WP_24	Systems & Technology	Future Architecture Requirements	Design a future-state platform architecture that enables federated, Fabric-style domain workspaces while maintaining enterprise guardrails and governance. Incorporate lessons and integrations from existing pilots, and define the uplift steps, security controls, integration patterns, and participation rules required for domains to operate safely, consistently, and at scale within the shared environment.	●	●	
WP_25	Systems & Technology	Automate Governance, Quality, and Lineage Controls	Introduce automated data governance practices, including data quality checks, lineage capture, metadata harvesting, access controls, and exception workflows.	●	●	
WP_26	Data Governance	Enterprise Semantic Layer & Modelling Standards	Develop a shared semantic layer that provides centrally governed, reusable business definitions and metrics that can be consumed by all domains and tools. This creates a single source of truth for KPIs, reduces duplicated modelling effort in spokes, and prevents inconsistent logic across reports. The semantic layer enables scalable self-service, supports automation, and becomes a key building block for federated analytics.	●	●	
WP_27	Culture & Adoption	Domain Data Community	Establish a community of practice for domain analysts, stewards and custodians to share patterns, learnings and improvement ideas. Define clear role expectations, progression levels and learning pathways to make data roles attractive and build capability across domains. Use this community to reinforce behaviours, uplift skills and continuously refine the hub-and-spoke model as it scales.	●	●	

Roadmap

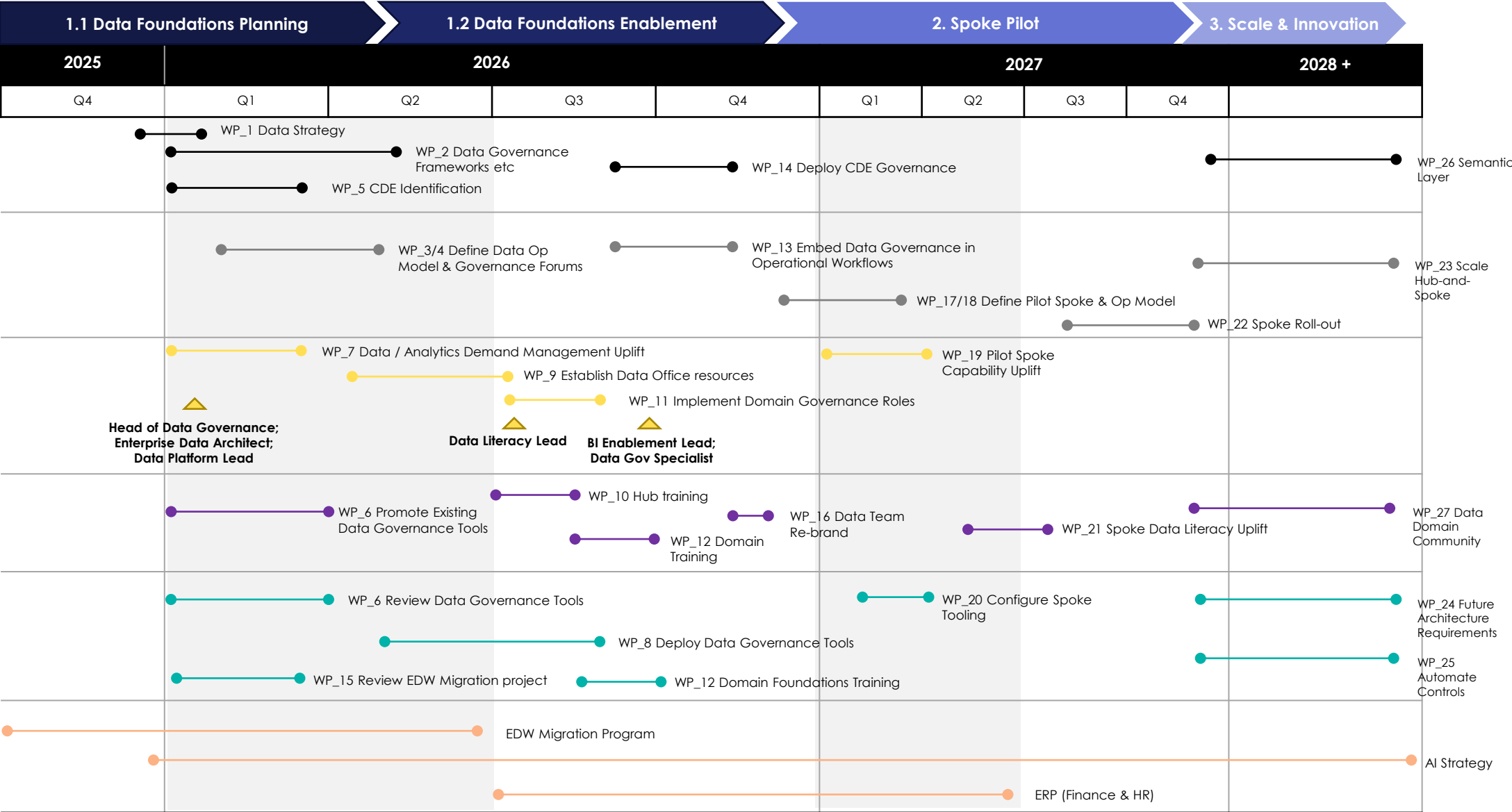
Key:

-  New / re-assigned role
-  Work package



Workstream

- Governance**
- Operating Model**
- Resourcing & Capability**
- Culture & Adoption**
- Systems & Technology**
- Key IT Initiatives**



Detailed Work Packages



Work Package 2 – Data Governance Frameworks, Guidelines & Policies



Work Package Summary

Develop and formalise a university-wide Data Governance Framework, including roles, responsibilities, standards, definitions, privacy-by-design principles, data quality rules, lifecycle management, and supporting governance policies.



Initiative Objectives

1. Establish a clear and consistent governance framework that defines roles, responsibilities, standards, and accountabilities across all domains.
2. Introduce & develop data governance policies, definition standards, and data quality controls to ensure safe, compliant, trustable data usage.
3. Provide a unified governance model and supporting guidelines that standardise data handling and reduce ambiguity across the university.



Key Activities

1. Assess existing governance artefacts, policies, and standards to determine what needs to be updated, consolidated, or newly developed.
2. Clarify which activities sit with IG (e.g., privacy, FOI compliance) and which sit with DG (e.g., data quality rules, definitions, lineage, lifecycle decisions), and outline how Risk and IT will interact. Capture any questions or follow-on actions or inclusions to the framework.
3. Define and document the Data Governance Framework, including roles, responsibilities, operating principles, lifecycle processes, and decision rights.
4. Develop supporting artefacts, such as definition standards, stewardship procedures, data quality rules, access standards, and privacy-by-design guidelines.
5. Validate framework components with key stakeholders and refine based on feedback and governance requirements.
6. Publish governance artefacts and socialise across domains through communication, training, and enablement activities.
7. Establish approval and change-control processes to manage updates and ensure ongoing governance sustainability.

Deliverables



- University-wide Data Governance Framework
- Role definitions (Owners, Stewards, Custodians, BSOs, Domain Leads)
- Governance standards (definitions, modelling, lifecycle, DQ, access, metadata)
- Developed data governance policies and guidelines
- Governance RACI and decision-rights matrix
- Stakeholder review pack and communication materials

Key Stakeholders



- SPP
- Information Governance
- Domain business leaders
- Data Stewards & Custodians

Key Risks



- Low adoption of governance standards across domains
- Conflicting legacy policies creating confusion
- Insufficient stakeholder engagement delaying finalization

Days estimate



- **Effort:** Medium
- **Days:** Approx 4 Months
- **External:** Recommended

Project Roles



- Associate Director, Strategy, Architecture & Data (Project Sponsor)
- BI&A (Responsible)

Dependencies



- Alignment with Hub and Spoke design
- Alignment with enabling technology (e.g. Databricks/Purview)
- Domain openness to take on data governance responsibilities
- Policy alignment to broader university frameworks

Work Package 3 – Data Operating Model Design

Work Package Summary

Define the target data operating model, detailing how the central hub, business domains, and governance roles work together. Establish key processes, responsibilities, and engagement pathways to enable consistent data delivery, governance, and scalable analytics and AI.

Initiative Objectives

1. Establish a clear, unified operating model that defines how domains interact with the Hub and how responsibilities are shared across data, governance, and analytics functions.
2. Provide standardised processes, handoffs, and service expectations to ensure data is produced, governed, and consumed reliably across the university.
3. Support scalable analytics and AI: Build a structure that supports certified datasets, domain data products, and safe federated analytics and AI development.

Key Activities

1. Define the Hub-and-Spoke structure including responsibilities, handoffs, domain participation rules, and governance interaction points.
2. Document roles and decision rights across the Hub, domain spokes, governance bodies, Stewards, Custodians, and business System Owners.
3. Establish core processes such as intake, prioritisation, dataset certification, modelling standards, quality management, metadata updates, and issue resolution.
4. Develop engagement pathways outlining how domains request services, contribute definitions, adopt standards, and interact with the Hub.
5. Define service catalogue and SLAs for Hub functions (engineering, modelling, certification, DQ, lineage, metadata).
6. Validate the model with key stakeholders across BI&A, SPP, IG, Records Management, and domain leaders.
7. Socialise and refine the model through communication, enablement, and early adoption activities.
8. Set up ongoing review and change-control mechanisms to evolve the model as new spokes onboard.

Deliverables

- Target Hub and Spoke Data Operating Model
- Roles, responsibilities, and decision-rights matrix
- End-to-end process definitions (intake, build, certify, maintain)
- Hub service catalogue and SLAs
- Domain engagement pathways and participation rules
- Operating Model overview pack for communication
- Change-control and continuous improvement plan

Key Stakeholders

- SPP
- Information Governance
- Domain business leaders
- Data Stewards & Custodians
- Cyber Security / Privacy (for guardrails)

Key Risks

- Lack of clarity on responsibilities delaying adoption
- Domains failing to adopt common processes or standards

Days estimate

- **Effort:** Medium
- **Days:** Approx 1-3 Months
- **External:** Recommended

Project Roles

- Associate Director, Strategy, Architecture & Data (Project Sponsor)
- BI&A (Responsible)

Dependencies

- Data Governance Framework & Policies
- Role clarity for Stewards, Custodians, System Owners – from guidelines

Work Package 15 – Review & Support EDW Migration Project



Work Package Summary

Review the existing EDW migration program to assess its alignment with the new Data Operating Model and governance structures. Identify impacts, risks, and dependencies on BAU reporting, downstream consumers, development of certified data assets, and the capacity of current teams. Ensure continuity of critical reporting, validate the fit of new datasets against emerging standards, and determine any required changes, resourcing needs, or adjustments to support successful integration into the future-state operating environment.



Objectives

1. Identify and mitigate risks to reporting, operations, and teams during the EDW migration.
2. Confirm the EDW program aligns with the new Data Operating Model, governance rules, and certification standards.
3. Understand reporting, dashboards, and consumer impacts to maintain business confidence and data reliability.
4. Ensure migrated datasets support certified data assets and meet enterprise definitions, quality rules, and modelling standards.
5. Identify workload pressure, capacity gaps, and additional resources required for safe delivery.



Key Activities

1. Review the EDW program status and completed work, including scope, design decisions, build progress, testing status, and alignment to current BI&A and SPP delivery patterns.
2. Perform a downstream reporting and consumer impact assessment, identifying which dashboards, reports, integrations, and stakeholders rely on affected datasets.
3. Conduct a full report inventory and rationalisation, mapping all dependent assets, identifying duplication, and determining where high-risk areas require prioritised attention.
4. Assess certified dataset development, confirming that new EDW datasets align with enterprise definitions, CDEs, modelling standards, lineage expectations, and governance guidelines.
5. Review the EDW project delivery schedule, ensuring coverage of key dates (enrolment cycles, finance periods, research deadlines, etc.) and establishing a priority sequence based on risk and business impact.
6. Evaluate impacts on the proposed Data Operating Model, including roles, handoffs, governance approvals, validation processes, and the fit of EDW workflows into the new ways of working.
7. Determine resourcing and uplift needs, assessing capacity gaps across BI&A and SPP to support BAU, EDW migration, dataset validation, and governance alignment simultaneously.

Deliverables



- EDW Impact Assessment Report (risks, dependencies, readiness)
- Downstream reporting & dashboard impact map
- Full report inventory, rationalisation and prioritisation list
- Certified dataset alignment review
- Updated migration priority schedule and risk mitigation plan
- Recommendations for changes to EDW scope, sequencing, or validation steps
- Resourcing requirements and uplift plan
- Integration guidance into new Data Operating Model & governance processes
- Stakeholder communication pack

Key Stakeholders



- BI&A
- SPP
- Information Governance
- Student, Finance, HR, and Research domain leads
- BI developers, analysts, dashboard/report owners
- Data Stewards & Custodians

Key Risks



- Misalignment between EDW migration and the new data operating model.
- Disruption to BAU data activities and services.

Days estimate



- **Effort:** Medium
- **Days:** Approx 1-2 Months
- **External:** Recommended

Project Roles



- Associate Director, Strategy, Architecture & Data (Project Sponsor)
- BI&A (Responsible)

Dependencies



- Data Governance Framework, Policies, and Standards
- CDE identification and definition lifecycle
- Data Operating Model Definition
- Metadata, lineage, and DQ tooling
- Data Office structure and resourcing uplift
- Reporting catalogue & downstream reporting inventory

Appendix

Appendix A. Use Case Deep Dive

Appendix B. Additional Information





Appendix A

Purpose of this section

Background

Sia's engagement centered on a single, high-value use case to anchor our findings and recommendations in a realistic example of how the university generates and relies on data. The student onboarding and enrolment journey was selected because it:

1. Is **strategically significant**: early-stage student retention is strongly dependent on a smooth onboarding experience, and the data captured during this phase underpins critical downstream reporting across the university.
2. **Spans multiple systems and functions**, making it an effective stress-test for cross-domain data, governance, and integration.
3. Currently experiences **substantial data quality issues**, with one in three students encountering an issue when enrolling online.

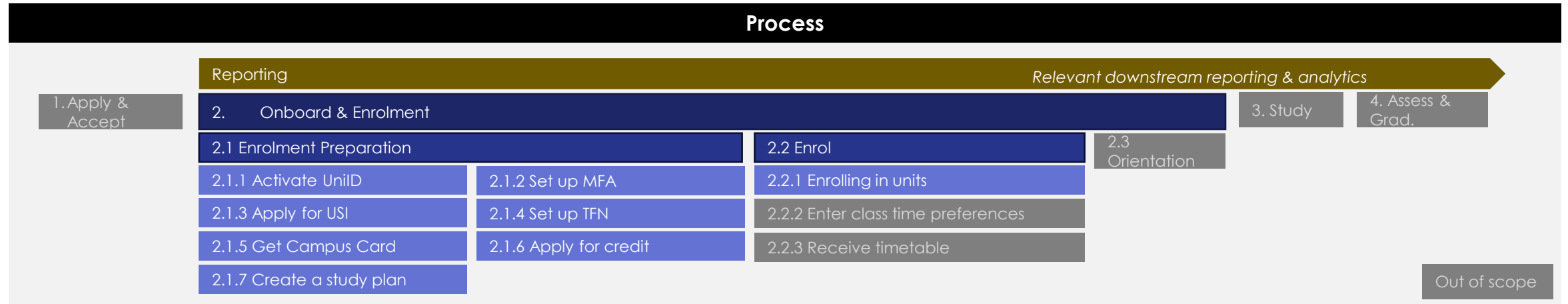
Purpose

This section illustrates the key issues identified within this use case and demonstrates how the proposed hub-and-spoke operating model and governance foundations will address and resolve these challenges.



Use Case Scope

Sia's engagement utilized key points in the **domestic, undergraduate student onboarding and enrolment process** to identify key pain points with the current data operating model and test the value / applicability of a target state operating model.



Teams

The in-scope process involves the following teams:

- DVCESE teams (e.g. Administration & Student Life) that work closely to move students along the onboarding and enrolment steps, providing advice (e.g. study plans) and trouble-shooting.
- DVCR (i.e. Library) helps students to resolve their IT issues so they can enrol successfully.
- SDVC (i.e. academic staff) are critical to ensuring accurate course data for enrolment and are often asked to review recognition of prior learning requests.

Key Systems

A range of internal, external and manual systems are used to coordinate this process, many with significant limitations that contribute to data quality issues. These limitations include:

- Multiple 'sources of truth' for student data with limited consolidation (e.g. Callista, Student Connect).
- No integration with external systems (e.g. eCAF, USI.gov), requiring high effort and high risk of human error to manually import data.
- Data is captured in 'local' systems with no integration (e.g. excel study plans, PDF credit plans).

While these system challenges require system uplift to resolve, the target data operating model can also support by establishing clear pathways and accountability to identify, prioritise, and escalate these issues for remediation.

Key Data Objects

Core data objects involved in this use case are:

Object	Description	Known Data Quality Issues
Student	Core entity capturing identity, contact details, and demographics.	Missing middle names; Manual check of DOB between ID & Callista; Manual check of photo against ID; Post code (?); incorrect phone numbers
Course / Curriculum	Defines program structure, majors/minors, unit rules, prerequisites.	Inconsistent course rules; outdated study plans; course changes not synchronised with enrolment
Credit / Advanced Standing	Captures recognition of prior learning and credit decisions.	No central precedent database of credit rules; missing lineage from Admissions
Enrolment	Records student participation in units for a semester.	Manual waivers without auditability

Use Case Objectives

Problem statement: The onboarding & enrolment process is fragmented across systems with inconsistent rules and weak data integration. This drives heavy manual effort for staff to address errors, impacts the student experience and erodes insight on student progression at a critical stage of the journey.

What problem are we trying to solve?

Core issues:

1. Inconsistent data entry due to system capability, user error and integration limitations.
2. Double entry & manual reconciliation across teams & systems.
3. Data captured outside systems (i.e. study plans) or unvalidated data (i.e. course requirements).

Impact:

1. Limited visibility of onboarding progress, both for operational staff and strategic decision makers.
2. Operational inefficiency and high manual effort for Student Admin and Student Life staff.
3. Negative student experience with inefficient and disconnected processes, delayed enrolment and (worst case) incorrect enrolment impacting entire UWA study experience.

What outcome are we aiming for?

Operational:

1. Reduction in manual fixes & exceptions
2. Improved visibility of individual student onboarding /enrolment progress to trouble shoot
3. Improved quality and connectivity of information & data

Strategic:

1. Improved student experience for domestic UG students across the onboarding process
2. Improved student & enrolment data for downstream reporting purposes (e.g. retention analysis)
3. Improved visibility of student attrition during onboarding/enrolment

How the Use Case Shaped the Roadmap

Many of the key pain points faced in the student onboarding and enrolment use case reflect enterprise-wide gaps in the university's data operating model and general maturity. The student onboarding journey was used as a practical lens to surface enterprise data issues, which directly informed the prioritisation and design of the roadmap initiatives.

No Data Governance

Data governance decisions are made after data is already in use, rather than when it is created or changed (e.g. enrolment rules clarified at reporting time; prerequisites enforced after students attempt enrolment).

Response

- Refreshing and operationalising the **Data Governance Framework**
- Developing and reinforcing **practical governance guidelines** (e.g. data ownership, stewardship)
- Ensuring governance is applied *at the point of data creation and change*, not just at reporting time

Inconsistent Data Definitions

The same student concepts are defined and used differently across systems and teams (e.g. credit decisions, legal name formats, bridging unit eligibility).

Response

- Define and prioritise student **Critical Data Elements (CDEs)**
- Establish clear ownership and quality accountability for each CDE
- Introduce consistent data quality standards and monitoring

Late Data Quality Detection

Errors are identified in dashboards or operational reports, requiring manual correction (e.g. invalid enrolments identified after census; reconciliation across reports).

Response

- Shift to **upstream validation** and (eventually) automated quality controls
- Progress an enterprise data quality assessment across all CDEs to support proactive data quality management

Unclear Analytics Ownership

No clear owner for student-domain insights, leading to reliance on central teams or ad-hoc local solutions (e.g. SPP producing faculty-specific dashboards; local Excel models in service teams).

Response

- Establish a hub-and-spoke operating model, **enabling the domain to create domain-specific analytics** within agreed guardrails.
- Clearly delineate enterprise responsibilities vs domain analytics ownership through **wide-spread education and training**.

Misused Analytics Capability

Strategic analytics capability (i.e. SPP) is consumed by routine reporting and dashboard requests (e.g. Power BI development for operational student reporting).

Response

- Position **SPP as the first spoke**, focused on strategic, cross-university student insights
- Enable domains to deliver their own reporting within clear guardrails

Capability gaps

Inconsistent understanding of roles, standards, and data practices leads to informal workarounds (e.g. spreadsheets maintained outside systems; reliance on individual knowledge).

Response

- Deliver **targeted change, training, and enablement** aligned to new roles
- Embed governance and data practices into day-to-day workflows



Appendix B

Data Governance Roles

Below outlines the proposed set of data governance roles in relation to the existing Information Management meeting. We will need to confirm the alignment and distinction between information governance roles and data governance roles and determine whether they should be managed separately. These will be addressed within the data governance roadmap activities.

Information Governance Role	Mapped Data Governance Role	Additional Data Governance Roles Needed	Combined Activities (Both IG + DG)	Data Governance-Specific Activities	Information Governance-Specific Activities
UWA Information Owner	CIO / Data Governance Lead	–	• Sets overall governance direction • Approves enterprise-wide governance framework • Provides executive escalation pathway • Ensures alignment of information & data governance	• Owns enterprise data & governance strategy • Sets data decision rights • Prioritises data governance roadmap • Oversees Data Office Hub	• Owns information policies (privacy, FOI, retention, classification) • Oversees information risk and compliance
Information Stewards	Data Stewards (Spokes)	• Data Stewardship Coordinator (Hub)	• Maintain definitions & rule clarity • Validate domain business meaning • Participate in governance forums • Raise DQ issues & approve resolutions	• Approve business definitions & KPI/business rule changes • Ensure alignment to enterprise glossary • Support dataset certification	• Ensure definitions align to information handling requirements • Ensure information access and privacy requirements are followed
Information Custodians	Domain Data Custodians (Spokes)	–	• Ensure correct data entry & source quality • Apply privacy and classification rules • Log data quality issues • Support remediation	• Apply enterprise DQ rules • Ensure lineage/metadata quality • Participate in DQ workflows • Enforce modelling/standard compliance at source	• Ensure information retention and disposal • Maintain compliance in records systems (TRIM/Content Manager)
Information Power Users	Domain Analysts / Power Users (Spokes)	–	• Support local reporting users • Apply governance guidelines in dashboards • Identify issues and escalate early	• Use certified datasets from CAF • Apply BI standards • Adopt KPI catalogue & shared measures • Provide requirements to CAF	• Reinforce information access policies • Ensure correct interpretation of sensitive information
Information Users	Data Consumers	–	• Enter and use data responsibly • Report issues • Follow usage guidelines	• Use certified datasets, not raw data • Follow data quality processes	• Follow information handling and FOI/privacy requirements
Business System Owner	IT Application Owner / System Owner	• Data Architect Lead (Hub) • Data Standards & Metadata Lead (Hub)	• Ensure systems support governance rules • Approve changes to data flows • Embed IG + DG requirements in system design	• Maintain metadata, lineage & integration standards • Ensure architecture aligns with domain boundaries • Implement data access/RLS controls	• Approve information classification & retention rules • Ensure systems comply with FOI, security, privacy policies
<i>N/A in IG model</i>	Enterprise Data Quality Lead (Hub)	(New Role or assigned to existing role)	• Shared DQ remediation with domains • Provide aggregated reporting to governance forums	• Define DQ rules, profiling, monitoring • Run DQ issue lifecycle • Drive DQ improvements across domains	• Ensure DQ supports compliance reporting (TEQSA, HEIMS, FOI accuracy)
<i>N/A in IG model</i>	Data Standards & Metadata Lead (Hub)	(New Role or assigned to existing role)	• Maintain enterprise metadata alignment • Coordinate definition agreements	• Maintain business glossary • Manage lineage, modelling standards • Oversee dataset certification	• Ensure classification rules and retention metadata are aligned
<i>N/A in IG model</i>	Data Architect Lead (Hub)	(New Role or assigned to existing role)	• Ensure architecture supports compliance + analytics	• Define data domains and federation guardrails • Approve integration patterns • Oversee semantic layer alignment	• Ensure system changes comply with IG policy (privacy, retention, access)
<i>N/A in IG model</i>	Analytics Engineer / CAF (Central Analytics Function)	(New Role or assigned to existing role)	• Build enterprise insights aligned to governance requirements	• Build certified datasets, semantic models, enterprise KPIs • Enforce BI standards • Support Spoke Analysts	• Ensure dashboards comply with information access rules

Example RACI

Within the development of data, analytics reports and other. Below outlines example RACI that would be further validated within the definition and deployment of the optimized data office and deployment of domain analytics within spokes.

Domain Analytics - (e.g., SPP): Builds domain-specific analytics using certified gold data.

Data Steward: Owns business meaning, definitions, rules, and CDEs.

BI Enablement Team: Governs enterprise-certified datasets.

Data Office – Data Engineers: Engineers ingestion, transformation, and platform operations.

Data Custodian: Governs access, security, classification, and system-level controls.

Activity	Analytics Spokes	Data Steward	BI Enablement Team	Data Office – Data Engineering	Data Custodian
Log request / requirement	R	C	C	I	I
Validate definitions & CDEs	C	A/R	C	I	C
Triage request (exists vs new)	C	C	A/R	R	I
Source classification	I	C	I	I	A/R
Bronze ingestion	I	I	I	A/R	C
Silver transformation	I	C	I	A/R	I
DQ checks & conformance	I	A/R	C	R	I
Build/Enhance Gold dataset	I	C	C	A/R	I
Certify dataset (governance forum)	I	A/R	R	I	C
Build IT Operational dashboards	I	C	A/R	I	I
Build domain dashboards	A/R	C	I	I	I
UAT & validation	A/R	R	C	I	I
Publish dataset/report	C	C	A/R	I	I
Ongoing DQ & lineage mgmt	I	A/R	R	R	C
Manage enhancements & lifecycle	C	A/R	R	I	I



Thanks

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